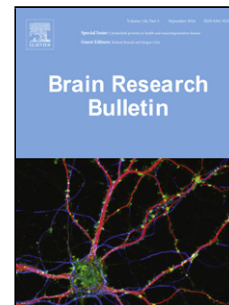


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iTRAQ-based proteomic analysis discovers potential biomarkers of diffuse axonal injury in rats

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Highlights:

- We focused on the corpus callosum in iTRAQ integrated with LC-MS/MS analysis.
- Among 514 DEPs, 14 common DEPs were present at all seven time points postinjury.
- PRPH and CSEN were successfully confirmed by western blotting and IHC analysis.
- PRPH and CSEN showed good potential as DAI biomarkers which need further study.

Abstract: Diffuse axonal injury (DAI) is one of the most common and severe pathological consequences of traumatic brain injury (TBI). The molecular mechanism of DAI is highly complicated and still elusive, yet a clear understanding is crucial for the diagnosis, treatment, and prognosis of DAI. In our study, we used rats to establish a DAI model and applied isobaric tags for relative and absolute quantitation (iTRAQ) coupled with liquid chromatography-tandem mass spectrometry (LC-MS/MS)

analysis to identify differentially expressed proteins (DEPs) in the corpus callosum. As a result, a total of 514 proteins showed differential expression between the injury groups and the control. Among these DEPs, 14 common DEPs were present at all seven time points postinjury (1, 3, 6, 12, 24, 48, and 72 h). Next, bioinformatic analysis was performed to elucidate the pathogenesis of DAI, which was found to possibly involve calcium ion-regulatory proteins (e.g., calsenilin and ryanodine receptor 2), cytoskeleton organization (e.g., peripherin, NFL, NFM, and NFH), apoptotic processes (e.g., calsenilin and protein kinase C delta type), and inflammatory response proteins (e.g., complement C3 and C-reactive protein). Moreover, peripherin and calsenilin were successfully confirmed by western blotting to be significantly upregulated during DAI, and immunohistochemical (IHC) analysis revealed that their expression increased and could be observed in axons after injury, thus indicating their potential as DAI biomarkers. Our experiments not only provide insight into the molecular mechanisms of axonal injury in rats during DAI but also give clinicians and pathologists important reference data for the diagnosis of DAI. Our findings may expand the list of DAI biomarkers and improve the postmortem diagnostic rate of DAI.

Keywords: Diffuse axonal injury; iTRAQ; proteomics; biomarkers; rat

1. Introduction:

Diffuse axonal injury (DAI), as one of the most common and severe pathological consequences of closed head injury or traumatic brain injury (TBI), is characterized mainly by widespread axonal damage in cerebral neural tracts in the corpus callosum and brainstem, with significant mortality as well as morbidity (Johnson et al., 2013; Smith et al., 2003). The pathogenesis of DAI is highly complicated, and its clear understanding is pivotal for an early diagnosis, appropriate therapeutic options, and accurate prognosis of patients with DAI. To date, numerous studies have indicated that DAI usually represents a progressive process related to calcium influx and overload, axonal cytoskeleton network degradation, mitochondrial damage, caspase-

mediated programmed cell death, and inflammatory responses (Buki et al., 2000; Siedler et al., 2014; Staal et al., 2010; Tang-Schomer et al., 2012). Due to the complicated pathogenesis underlying DAI, there is no uniform standard for the diagnosis (Ma et al., 2016). In the clinic, most commonly used diagnostic methods are noninvasive, such as diffusion-weighted imaging and diffusion tensor imaging, which are more sensitive than computed tomography in the detection of DAI (Chang et al., 2010; Ezaki et al., 2006). Nonetheless, inherent variation in the characteristics of patients with TBI and differences in data acquisition, analytical methods, and costs still impede generalized conclusions about the practical utility of diffusion-weighted imaging and diffusion tensor imaging in DAI (Hulkower et al., 2013; Tsitsopoulos et al., 2017). In histopathological analysis, β -amyloid precursor protein (β -APP) immunocytochemistry remains the gold standard for assessing the presence and severity of traumatic axonal injury (Johnson et al., 2013). It is worth noting that accumulation of β -APP in axons is not exclusive to trauma and has been detected after other problems including ischemia, hypoxia, and hypoglycemia (Dolinak et al., 2000; Kaur et al., 1999; Oehmichen et al., 1999). Because each individual method of diagnosing DAI has specific limitations, the search for more reliable biomarkers has been attempted as an alternative way to diagnose and quantify DAI.

Currently, the field of research on DAI-specific diagnostic and theranostic biomarkers is expanding, and considerable progress has been made in recent years (Ma et al., 2016). In addition to β -APP, commonly used biomarkers for the diagnosis of acute DAI and for analysis and prognosis of the conditions associated with DAI include spectrin-II subunit and its degradation products SBDP145, SBDP150, and SNTF (Johnson et al., 2016; Reeves et al., 2010). Other biomarkers include neurofilaments (NFs), tau protein, and myelin basic protein as well as inflammatory mediators, such as IL-1 β , IL-6, and TNF- α (Cengiz et al., 2008; Lin et al., 2013; Ljungqvist et al., 2017; Winkler et al., 2003). Nonetheless, these biomarkers commonly have different specificity and sensitivity as well as limited availability for medical tests and use in daily practice. Therefore, the quest for DAI biomarkers for early diagnosis and a deeper understanding of the molecular mechanism of DAI has to

be continued.

Recently, with rapid advancement of proteomics-based techniques, the development of isobaric tags for relative and absolute quantitation (iTRAQ)-based nano-liquid chromatography-tandem mass spectrometry (nano-LC–MS/MS) by Darryl Pappin and colleagues at Applied Biosystems (Ross et al., 2004) led to the widespread and successful use of this method for biomarker discovery in brain injury and various diseases (Crawford et al., 2012; Datta et al., 2013; Ren et al., 2016). This approach has some advantages: it allows for simultaneous comparison of protein profiles in complex biological samples, thereby providing information on the identification and relative quantification of peptides and proteins, and this method can be enhanced by means of powerful bioinformatic tools and statistical analysis (Muth et al., 2010; Schwacke et al., 2009).

So far, the pathogenesis of DAI has not been comprehensively elucidated, especially at the level of the proteome. Therefore, an iTRAQ-based quantitative proteomic approach may provide new information pertaining to the progression of DAI. In this study, we established a rat model of DAI and identified differentially expressed proteins (DEPs) in the corpus callosum tissues by iTRAQ integrated with LC-MS/MS analysis. The possible biological roles of these DEPs in DAI progression were next evaluated by various bioinformatic analyses. These findings may not only provide new insights into the pathophysiology and pathogenesis of DAI but also give clinicians and pathologists several early theranostic and prognostic biomarkers of DAI.

2. Experimental procedures

2.1. *The animal model*

All animal experiments were approved and monitored by the Animal Ethics Committee of Huazhong University of Science and Technology. All the animals were purchased from the experimental animal center of Tongji Medical College, Huazhong

University of Science and Technology.

Female Sprague-Dawley (SD) rats weighing between 270 and 300 g (10-12 weeks old) were housed in a controlled-temperature environment on a standardized light/dark cycle (12 h/12 h), with uninterrupted access to food and water. The rats were randomly assigned to a sham group and DAI groups. Rats of DAI groups were subjected to a diffuse traumatic axonal injury by the weight drop impact acceleration procedure based on the Marmarou method (Foda et al., 1994; Marmarou et al., 1994). General anesthesia was induced by inhalation of 5% isoflurane. Briefly, after a midline incision between coronal and lambdoid sutures, a steel disk (10 mm in diameter and 3 mm thick) was adhered to the skull between the bregma and lambda suture. Each rat was placed on a foam bed in the prone position and a 450 g steel cylinder was then dropped from 2 m height within a vertical tube positioned directly over the steel disc on the rat's head. Then, the rat was immediately removed so that any subsequent impacts due to weight bouncing were prevented to ensure only one impact on the disc. The scalp was sutured after gentle removal of the steel disk from the skull. In the control (sham-injured) group, the rats underwent the same surgical procedure without the impact. The rats were euthanized at 1 h, 3 h, 6 h, 12 h, 24 h, 48 h, or 72 h postinjury and were compared to the sham animals euthanized at 72 h post-sham injury. For experiments with iTRAQ and histopathological examination, 10 rats and 5 rats were included in each group, respectively. For confirmation via western blotting, 3 rats were included in each group (Table 1).

2.2. Histopathological analysis

To verify whether the DAI animal model is successful, paraffin-embedded cerebral tissues (five rats were included in each DAI group and sham group) were subjected to routine histopathological analysis including hematoxylin and eosin (H&E) staining, silver staining, and immunohistochemical (IHC) analysis of β -APP as described below. All tissue examinations were performed on 4 μ m sections.

IHC analysis of β -APP was performed on the formalin-fixed cerebral tissue

sections according to standard protocols. The sections were dewaxed and then rehydrated in a series of ethanol solutions, followed by high-temperature antigen retrieval via microwaving in Tris–ethylenediaminetetraacetic acid (EDTA) buffer (10 mM Tris, 1 mM EDTA, pH 9.0) for 20 min. Endogenous peroxidase was blocked on each section by incubation for 15 min with 0.03% hydrogen peroxide. After blockage in 5% normal goat serum at room temperature for 30 min, a primary antibody (anti- β -APP, ab105122, Abcam) was applied at a 1:2000 dilution in an antibody diluent at 4°C overnight. The following day, after washing, the sections were incubated with a horseradish peroxidase–conjugated secondary antibody for 30 min at room temperature. Finally, all the slides were incubated for 1–2 min with 3-3'-diaminobenzidine, counterstained with hematoxylin, clarified, and mounted. Negative control was set up by replacing the primary antibody with the antibody diluent. All the slides were examined under a Nikon Eclipse i-S microscope (Tokyo, Japan).

2.3. Quantitative proteomic analysis

2.3.1 iTRAQ sample preparation

For the iTRAQ analysis, in each group, fresh corpus callosum tissues were rapidly dissected from the brains and sampled. Corpus callosum region was identified based on Paxinos and Watson stereotaxis brain atlas (Paxinos et al., 2006), and the method to dissect corpus callosum was as follow: Following deep anesthesia and transcardial perfusion, brain was removed from the skull immediately and placed on the filter paper at –80 °C for 40–50 s. Then brain was placed dorsal side up and the cerebral cortex was carefully stripped laterally along the sagittal midline under a stereomicroscope. After that, corpus callosum region (coronal bregma -5.5 to +1.5) was exposed and the whole corpus callosum was carefully dissected. All of the dissected procedures were performed on ice. Immediately after that, the tissue samples were snap-frozen in liquid nitrogen and stored at –80 °C until further processing. The remaining samples were formalin fixed and subjected to IHC analysis and verification of iTRAQ-identified candidate proteins. Within each group, 10 corpora callosa were pooled: sham (n = 10) and rats that were euthanized at 1 h (n =

10), 3 h (n = 10), 6 h (n = 10), 12 h (n = 10), 24 h (n = 10), 48 h (n = 10), or 72 h (n = 10) postinjury, to ensure sufficient amounts of tissue samples for the proteomic analysis.

2.3.2 Protein extraction and iTRAQ labeling

Samples from the eight groups were prepared by the liquid-nitrogen grinding method. Next, 200 μ l of radioimmunoprecipitation assay (RIPA) lysis buffer (50mMTris-HCl, 150mMNaCl, 1%SDS, 0.1%TritonX-100, pH 8.0) was added to each sample. After the samples dissolved, the lysates were sonicated for 1 min (20 W, a pulse at 10 s then a 15 s pause for 10 cycles), and cell debris were removed by centrifugation at 4 °C and 12,000 rpm for 20 min. The supernatant was precipitated for 2 h with cold acetone at -20 °C, followed by centrifugation at $12,000 \times g$ for 20 min. After the supernatant was discarded, 200 μ l of RIPA lysis buffer supplemented with 1 nM phenylmethylsulfonyl fluoride (PMSF) and 2 mM EDTA was added to the resulting pellet. After 5 min, 10 mM dithiothreitol (DTT) was added followed by sonication and centrifugation. After that, the supernatant was incubated at 56 °C for 1 h with a DTT solution at a final concentration of 10 mM, and 55 mM iodoacetamide (IAA) was then added, followed by incubation for 45 min in darkness. Next, the supernatant was precipitated for 2 h with cold acetone at -20 °C, followed by centrifugation again. After that, the precipitate was dissolved in 200 μ l of 0.5 M tetraethylammonium bromide (TEAB) and sonicated, then centrifuged simultaneously under the same conditions. The supernatant was collected and transferred to a fresh tube. The supernatant was pipetted for determination of protein concentration by the Bradford assay (Thermo Fischer Scientific, USA).

An aliquot of total protein (200 μ g) was taken from each sample solution and mixed with 4 μ l of the reducing reagent and incubated at 60 °C for 1 h. Next, 2 μ l of the cysteine-blocking reagent was added and incubated for 10 min at room temperature, and then centrifuged at $12,000 \times g$ for 20 min. The precipitate was then removed and reconstituted with 100 μ l of 1 M TEAB, followed by centrifugation; the

bottom phase of the tube was collected, and the whole procedure was repeated three times. Then, the protein samples were digested with trypsin (Promega, WI, USA) overnight at 37 °C, at a ratio of protein to trypsin of 50:1. The next day, the peptides were centrifuged at $12,000 \times g$ for 20 min, and the bottom phase of the tube was collected. After that, 50 μ l of 1 M TEAB was added, and the mixture was centrifuged under the same conditions. In combination with the previous step, 100 μ l samples were obtained at the bottom of the tube.

Then, the samples were labeled with 8-plex iTRAQ reagents according to the manufacturer's protocol (AB Sciex, MA, USA) as follows: 131 for the sham group and 114, 115, 116, 117, 118, 119, and 121 iTRAQ tags for samples corresponding to time points 1, 3, 6, 12, 24, 48, and 72 h postinjury, respectively. Then, the contents of each labeled sample were mixed, incubated (room temperature, 2 h), and dried before further analyses.

2.3.3 High-pH reverse-phase liquid chromatography fractionation

The labeled samples were first separated on a high-pH reverse-phase liquid chromatography column (Dionex Ultimate 3000, USA). The labeled samples were reconstituted in mobile phase A (20 mM HCOONH₄, pH 10) and separated on an analytical column (Gemini-NX 3u C18 110A; 150*2.00 mm) with a solvent gradient of 5–90% Buffer B (20 mM HCOONH₄, 80% ACN, pH 10) in mobile phase A, at a flow rate of 200 μ l/min. Total gradient duration was 90 min. Absorbance at 214 nm was monitored, and a total of 24 fractions were collected and dried in a vacuum concentrator for subsequent experiments.

2.3.4 LC-MS/MS analysis

Each fraction was reconstituted in 0.1% formic acid in 2% ACN for LC-MS/MS analysis on a Q Exactive mass spectrometer (Thermo Fisher Scientific) coupled to an UltiMate 3000 HPLC system (Dionex LC Packings). The peptide mixture was initially separated on a 10 cm C18 reversed-phase column (75 $\times$ 150 mm, 2 μ m, 100

Å) at a constant flow rate of 300 nl/min with a linear 65 min gradient, which consisted of mobile phase A (0.1% formic acid) and mobile phase B (0.1% formic acid in 80% acetonitrile). After sample injection, the column was equilibrated with 4% mobile phase B for 5 min, and the following gradient schedule was then initiated: from minute 5 to minute 45, the mobile phase B concentration was increased to 50%; during minutes 45–50, mobile phase B concentration was increased to 90%; and maintained at 4% during minutes 50.2–65. The eluted peptides were analyzed on the mass spectrometer. The following MS parameters were set: mass range, 350–1800 m/z; resolution, 70,000; AGC target, 3e6; and maximum IT, 40 ms. The following parameters of the MS/MS were chosen: resolution, 17, 500; AGC target, 1e5; maximum IT, 60 ms; TopN, 20; and NCE/stepped NCE, 30. To minimize the effects of experimental variation, three independent MS/MS runs were performed on each sample.

2.3.5 Data analysis

Protein identification and quantification were performed in ProteinPilot™ 4.5 software (AB Sciex). Searches were based on the Paragon database search algorithm and an integrated false discovery rate analysis function for peptide identification (Lam et al., 2010; Shilov et al., 2007). We retrieved the MS/MS spectra from Rat_Uniprot_2015.3.2, which covers 32,983 of protein sequences. According to the standard parameters, a unique protein had to contain at least two unique peptides, and the false discovery rate had to be less than 1%. To strengthen credibility, we considered only the iTRAQ ratio of the proteins within the range 0.5–20. A 95% confidence interval was chosen as the significance threshold for protein identification, and the detected protein threshold (unused ProtScore) was set to ≥ 1.3 to minimize the false positive identification of proteins.

Functional classification and Gene Ontology (GO) enrichment analyses of the DEPs were carried out using DAVID (<https://david.ncifcrf.gov/>). Proteins were classified by GO category (<http://www.geneontology.org>), including “biological

process,” “cell component,” and “molecular function.” The KEGG (<http://www.genome.jp/kegg/>) database was employed to identify significantly enriched pathways. *Rattus norvegicus* was selected as the species and the background. The significance was determined with slight modifications as recommended by the authors of DAVID according to the Benjamini-corrected P value <0.05. Functional protein association networks were explored in STRING v.10.5 (<http://string-db.org/>). The heat map was generated in MeV 4.9 software.

2.4. Western blotting

In order to verify the results of the proteomic analysis, western blotting of two representative DEPs, peripherin (PRPH) and calsenilin (CSEN) was performed. Additionally, α -II-spectrin breakdown products (SBDPs) of 150 kDa (SBDP150) was applied to demonstrate the role of calpain-dependent proteolysis in DAI. The corpus callosum samples (40 μ g of total protein) from DAI groups and the sham group (three in each group) were separated by SDS-PAGE on a 10% gel and transferred to polyvinylidene difluoride (PVDF) membranes (0.45 m, Millipore). After blockage with 5% nonfat milk at room temperature for 2 h, the membranes were incubated with the specific primary antibodies—rabbit anti-PRPH (1:1000 dilution, ab99942, Abcam), rabbit anti-CSEN (1:1000, ab99043, Abcam), rabbit anti-SBDP150 (1:10000, ab133320, Abcam), and anti-GAPDH (1:5000, ab8245, Abcam)—overnight at 4 °C, followed by incubation with the horseradish peroxidase–conjugated secondary antibody (1 : 5000, BOSTER) for 1 h at 37°C. Finally, the proteins on PVDF membranes were visualized via enhanced chemiluminescence, and protein quantification was performed in the ImageJ software (<http://rsb.info.nih.gov/ij/>). The results were normalized to GAPDH; three independent experiments were conducted for each data point, and the data were statically analyzed by one-way analysis of variance (ANOVA) followed by the post hoc Dunnett’s T3 multiple comparison test in SPSS 18.0 (SPSS Inc., Chicago, IL, USA). Significance was considered when the p-value <0.05.

2.5. Immunohistochemistry

IHC analysis of PRPH and CSEN was performed on the formalin-fixed cerebral tissue sections according to standard protocols (details as IHC analysis of β -APP in section 2.2). Briefly, after dewaxing, dehydration, antigen retrieval and blockage, the sections were randomly selected for incubation with appropriate primary antibodies against PRPH (1:1200, ab4666, Abcam) and CSEN (1:200, ab3474, Abcam) in an antibody diluent at 4°C overnight. The following day, after secondary antibody incubation, all the slides were incubated with 3-3-diaminobenzidine, counterstained with hematoxylin, clarified, mounted and examined under microscope.

3. Results

3.1. Pathological changes during DAI in the rats

To evaluate our DAI model, H&E staining, silver staining, and β -APP IHC analysis were carried out to effectively demonstrate changes associated with the axonal pathology. On H&E-stained sections, neuronal pyknosis, swelling, torsion, and extracellular space expansion were observed in the DAI groups. Both silver staining and β -APP IHC analysis visualized abnormal axons in the DAI groups. Silver-stained sections in the sham group revealed smooth and continuous axons with uniform thickness that were arranged closely. By contrast, in the DAI groups, we noticed swollen and clearly twisted axons, even axonal rupture and formation of axonal retraction balls, which were observed at all time points from 3 to 72 h postinjury, and the magnitude of these signs markedly increased from 24 to 72 h postinjury (Fig. 1). The axonal pathology was confirmed by β -APP IHC staining, which is an excellent indicator of axonal injury (Hortobagyi et al., 2007; Johnson et al., 2013; McKenzie et al., 1996; Sherriff et al., 1994). Increased numbers of β -APP-positive axons were observed at all time points during DAI. The β -APP-positive axons had varicose or beadlike morphology, often with multiple swollen regions along the axon length, and greatly increased in diameter. In particular, many axonal β -APP positivity patterns

had the appearance of axonal retraction balls at disconnected axonal terminals. Significant increases in the numbers of β -APP-reactive axonal varicosities and retraction balls were detected in the corpus callosum and brainstem at 24–72 h postinjury. The above-mentioned β -APP-reactive axons with abnormal morphology were seen in the subcortical and periventricular white matter, corpus callosum, and brainstem. No β -APP-stained axons were visible in the sham rats. In our study, we focused on the axonal pathology in the corpus callosum (Fig. 2). These results confirmed that the DAI model analyzed in this study is reliable and was suitable for the subsequent experiment.

3.2. Quantitative proteomics by iTRAQ coupled with LC-MS/MS

To identify the DEPs associated with the pathophysiology and pathogenesis of DAI, we compared protein expression of each DAI group with that in the sham group. A total of 4460 proteins were identified and quantified overall in the corpus callosum. The labeling efficiency was 95.7%. Proteins featuring a fold change >2.0 -fold or <0.5 -fold were regarded as differentially expressed. Through comparative analysis, 514 DEPs in at least one time point group were identified and quantified as compared to the sham group. Supplementary Table A describes the proteins in detail. We found 71, 74, 77, 63, 56, 56, and 63 upregulated proteins and 159, 122, 63, 88, 83, 48, and 100 downregulated proteins at 1, 3, 6, 12, 24, 48, and 72 h postinjury, respectively. The top 10 upregulated and downregulated DEPs identified in DAI samples at each time point are presented in Fig. 3. Our proteomic analysis was focused on the common DEPs detected at all seven time points postinjury; this set contained 14 proteins, with 10 upregulated and four downregulated ones (Table 2). We observed that KANK1, PRPH, and KCNIP3 manifested a more significantly upregulated expression pattern than did other proteins. Then, we classified these common DEPs by similar expression trends, and this analysis revealed that Col6a1 and Th had a similar expression trend, as did the group Myef2, Prkcd, Adam11, and Plcl2.

3.3. *Bioinformatics analysis*

To gain insights into the functions that may be affected by DAI, the DEPs were classified by bioinformatic analysis. They were categorized according to their GO terms for biological processes, molecular functions, and cellular components (Fig. 4A–C). The most abundant proteins in the biological process category were related to “response to organic substance” (13.8%), “response to endogenous stimulus” (10.6%), and “ion transport” (10.6%), followed by intracellular signaling cascade (10.3%) and macromolecular complex assembly (9.6%; Fig. 4A). For the molecular-function category, Fig. 4B shows that the most represented function was ion binding at 20.2%, followed by nucleotide binding (16.1%), ribonucleotide binding (13.9%), ATP binding (10.7%), calcium ion binding (8.9%), and structural molecule activity (7.7%). In the cellular-component category, the majority of proteins identified (15.3%) were associated with the plasma membrane. Other significant components included cytosol, mitochondrion, non-membrane-bounded organelle, cell fraction, extracellular region, organelle membrane, membrane-enclosed lumen, and cytoskeleton (Fig. 4C).

In addition, by means of the KEGG pathway database and DAVID, these DEPs were mapped to 28 pathways in total. The most enriched pathways included Alzheimer’s disease, Huntington’s disease, Parkinson’s disease, oxidative phosphorylation, and calcium signaling pathway (Fig. 5). To determine the key proteins in the function network for the 514 DEPs, online tool STRING v.10.5 was applied to reveal protein–protein interactions and their network (Jensen et al., 2009). The results uncovered six clusters determined by the K-means method, represented by nodes with different colors (Fig. 6), which revealed that most of the DEPs are tightly related in a network, and several proteins like Pkm, Gnb1, Acly and Calm1 were found to be located in the connected nodes of the protein functional interaction network.

Hierarchical clustering was conducted in MeV 4.9 software to visualize the changes in abundance of all the top 10 upregulated and downregulated DEPs at each

time point postinjury (Fig. 7). The heat map indicated that there were fewer upregulated proteins than downregulated ones among the 79 DEPs. The changes in protein abundance in the 3 h group were significantly different from those found in the other six groups. The relative abundance patterns of the proteins in the 1 h and 6 h groups were similar, pointing to a similar pathological process at 1 and 6 h postinjury, as was the case for the 12 h and 48 h groups as well as the 24 h and 72 h groups.

The main pathological mechanisms of DAI are related to calcium influx and overload, degradation of the axonal cytoskeleton network, mitochondrial damage, caspase-mediated programmed cell death, and inflammatory responses (Buki et al., 2000; Siedler et al., 2014; Staal et al., 2010; Tang-Schomer et al., 2012). Thus, to gain further insight into the biological processes that respond to DAI, we focused only on proteins related to cytoskeleton organization, calcium ion regulation, an apoptotic process, and an inflammatory response (Table 3).

3.4. Validation of iTRAQ-based proteomics results for selected DEPs

Two upregulated proteins, PRPH (gene name: Prph) and CSEN (gene name: Kcnp3), were selected for confirmation via western blotting and IHC analysis on the basis of their biological functions (degradation of the axonal cytoskeleton network, calcium ion regulation, and an apoptotic process) in axonal injury during DAI and their large fold changes. Western blotting results revealed that PRPH was significantly upregulated ($p < 0.05$) at all the time points during DAI relative to the sham group. These data revealed an expression pattern similar to the results of iTRAQ analysis. Similarly, the KCNIP3 amount significantly increased ($p < 0.05$) in the DAI groups except at 6 and 24 h postinjury ($p > 0.05$) relative to the sham group (Fig. 8).

As we all know, the corpus callosum not only contains axons but neurogliaocytes (oligodendrocytes, microglia, and astrocytes). Thus, to ascertain the cell type in which PRPH and CSEN are localized, IHC staining was performed. The PRPH-reactive profiles were mainly found in axons, and its reactive axons in the sham rat corpus callosum was very few with normal morphology, as demonstrated by IHC

staining (Fig. 9). Following DAI, the number of PRPH-reactive axons was increased in all areas of the corpus callosum, and most axons displayed altered morphologies (undulations, swellings, degeneration). It is worth noting that PRPH-reactive axons were frequently of much smaller diameter than typically observed with β -APP IHC. The expression of CSEN in the sham rat corpus callosum was low and mainly detected in neurogliaocytes. Following DAI, CSEN immunoreactivity was more intense than in the sham group and punctate CSEN-reactive distributions were observed in some axons. The above expression patterns of CSEN were more prominent until 72 h postinjury (Fig. 9).

On the other hand, in addition to DEPs identified in iTRAQ, SBDP150, as one of the existing biomarkers of axonal injury (Manivannan et al., 2018), was also detected using western blotting. The result showed a significant increase at all the times postinjury relative to the sham group (Fig. 10), suggested α -II spectrin degradation induced by calpain activation (Valiyaveetil et al., 2014).

4. Discussion

DAI, as one of the most common and pivotal pathological features of TBI, is characterized histologically by widespread damage to the axons of the brainstem, corpus callosum, parasagittal white matter, and the gray-white matter junctions of the cerebral cortex (Johnson et al., 2013; Meythaler et al., 2001). In our study, mapping of β -APP-reactive axons onto abnormal morphology was consistent with the known pattern of DAI, thereby confirming that the DAI model in this study is reliable and suitable for the subsequent screening of DEPs.

The corpus callosum is a region of midline structures that is particularly vulnerable to shearing forces or rotation of the brain owing to trauma, and may serve as an indicator of poor prognosis (Cicuendez et al., 2017; Ubukata et al., 2016). Additionally, the corpus callosum is rich in axonal structures; accordingly, we had the opportunity to find specific biomarkers enriched in the axons via our quantitative proteomic study. Therefore, we focused on the corpus callosum in our study.

By iTRAQ-based quantitative proteomic analysis, we identified 514 proteins (in the DAI groups) that were differentially expressed relative to the control.

Bioinformatic analysis revealed that these 514 DEPs are involved in a multitude of biological processes including energy metabolism and many neurodegenerative diseases. Among them, 14 common DEPs were detected at all seven time points postinjury and might serve as confirmatory indicators of DAI that are involved in the generalized response to DAI.

Although the pathogenesis of DAI has not been comprehensively elucidated to date, a number of studies have suggested that DAI usually is a progressive process that is related to calcium influx and overload, degradation of the axonal cytoskeleton network, mitochondrial damage, caspase-mediated programmed cell death, and inflammatory responses (Buki et al., 2000; Siedler et al., 2014; Staal et al., 2010; Tang-Schomer et al., 2012). Therefore, in our study, we compiled a list of most of our identified DEPs—those that are involved in calcium ion regulation, cytoskeleton organization, apoptotic process, and inflammatory response—to make the discussion more informative and targeted as well as to find potential DAI biomarkers (Table 3).

4.1 Calcium mediates an injury cascade in DAI

To date, it has been demonstrated that DAI is a progressive event gradually evolving from primary axonal injury (the immediate axonal alteration caused directly by a mechanical insult) to secondary axonal injury due to a progressive molecular and cellular cascade of pathological changes within the axon after the initial mechanical insult (Buki et al., 2006; Johnson et al., 2013). Disruption of calcium homeostasis in the axoplasm is thought to be crucial for the secondary axonal injury as well as for delayed cell death or dysfunction after a mechanical deformation. After external instant shear and tensile forces, permeability of the axolemma changes, and a huge amount of calcium ions (Ca^{2+}) enters axons (Frati et al., 2017; Ma et al., 2016). Overactivation of calpain, a Ca^{2+} -activated neutral protease, due to high intracellular Ca^{2+} concentration is a fundamental event in the DAI-associated pathological changes and damages various cytoskeletal proteins such as microtubule-associated proteins,

spectrin, and NFs and initiates mitochondrial injury that results in the release of cytochrome c with activation of caspases (Buki and Povlishock, 2006; Liu et al., 2008). α -II spectrin (280 kDa), a major component of the cytoskeletal structure in neurons, is a substrate for calpain and caspase-3 and can be cleaved after axonal injury to generate α -II spectrin breakdown products (SBDPs), with molecular weights of SBDP145 and SBDP150 and SBDP120 (Czogalla et al., 2005; Goodman et al., 1995; Valiyaveetil et al., 2014). SBDPs are a consequence of calpain activation by Ca^{2+} influx in traumatic axonal injury and have been considered as one of the most prominent biomarkers of axonal injury (Manivannan et al., 2018). In this respect, SBDP150 was detected using western blotting in our study and the result showed significant increased levels of it as early as 1 h postinjury and remained elevated up to 72 h postinjury, indicating early Ca^{2+} -induced calpain activation in DAI. In iTRAQ analysis, we found 56 DEPs, such as *Plcb1*, *Ryr2*, *Prkcd* and *Kcnip3*, are associated with calcium ion regulation. Among these, CSEN (gene name: *Kcnip3*) is one of 14 common DEPs with significantly high fold changes at all the time points postinjury. CSEN, a neuronal Ca^{2+} -binding protein and a member of the recoverin family, serves as a calcium-regulated transcriptional repressor (Carrion et al., 1999), interacts with Alzheimer's disease-associated presenilin (Buxbaum et al., 1998), and binds to A-type potassium channels to control neuronal excitability in response to changes in intracellular calcium concentration (An et al., 2000; Jo et al., 2005). Some studies suggest that CSEN makes cells more susceptible to apoptotic triggers, possibly by elevating caspase 3 and calpain activities (Choi et al., 2001; Lilliehook et al., 2002), and the proapoptotic activity of CSEN is coordinated with presenilin or γ -secretase activity to play a crucial role in the neuronal death in Alzheimer's disease (Jo et al., 2005). In our study, the upregulation of CSEN at all the time points postinjury, as successfully confirmed by western blotting, may also be related to postinjury Ca^{2+} homeostasis or modulation and can be attributed to an apoptotic process. These findings suggest that inhibiting the expression of CSEN after an axonal injury may reduce Ca^{2+} overload-mediated neuronal death and degradation of the axonal cytoskeleton network and thus improve the clinical outcome. In addition, it has been

reported that CSEN is primarily expressed in neurons of the cerebellum and hippocampus, as well as in a subset of dendrites and axons (Hammond et al., 2003). Under certain pathological conditions, CSEN was also detected in reactive astrocytes (Jin et al., 2005; Larrode et al., 2018). According to our IHC analysis, we found that CSEN immunoreactivity was observed in neurogliocytes and axons in the corpus callosum following DAI, indicating that trauma may induce CSEN expression in areas where it is usually silent.

4.2 Axonal cytoskeleton alterations and breakdown

Regardless of the initial triggering process, the ensuing changes during DAI inevitably include axonal cytoskeleton alterations and breakdown, which are associated with post-traumatic increases in intracellular calcium concentration (Buki and Povlishock, 2006; McCracken et al., 1999). Elevated calcium concentration can activate calcium-dependent calpains, caspases, and phosphatases, all of which mediate cytoskeletal breakdown, leading to defective axonal transport, axonal swelling, and neurofilament (NF) compaction and disruption (Siedler et al., 2014; Smith et al., 2000). NFs are the dominant intermediate filaments in neurons and are a major component of the axonal cytoskeleton, whose elements may be important contributors to the tensile strength of axons and resilience to mechanical stretching (Lepinoux-Chambaud et al., 2013; Perrot et al., 2008; Siedler et al., 2014). They are composed of three types of polypeptides, which cause their categorization into subunits of different molecular mass: neurofilament light (NFL; 68–70 kDa), medium (NFM; 145–160 kDa), and heavy (NFH; 200–220 kDa) (Kobek et al., 2014). So far, the majority of western blotting analyses and immunohistochemical studies on animal experimental models and human material have demonstrated progressive loss of the cytoskeletal NF proteins after TBI, but a significantly elevated levels of SBDPs, emphasizing a role for calpain-mediated proteolysis in cytoskeletal proteins following TBI (Kupina et al., 2002; McCracken et al., 1999; Posmantur et al., 1994). These findings are in very close agreement with our data in the current study. According to our iTRAQ data, we found an acute and progressive loss of NF proteins following DAI,

manifesting as a significant decrease in the amounts of all the NF polypeptides—NFL, NFM, and NFH—at most of the time points postinjury (Table 3). Meanwhile, a significant increase in the levels of SBDP150 was detected by western blotting in our study, suggesting calpain activation may contribute to the degradation of NF proteins after DAI. Moreover, we found that more DEPs related to cytoskeleton organization were downregulated, indicating that the axonal-cytoskeleton breakdown plays a part in the development and progression of DAI-associated pathologies.

Of note, our data revealed that several DEPs involved in cytoskeleton organization were significantly upregulated, especially PRPH, which is one of the common DEPs detected at all the seven time points postinjury, implying that PRPH holds promise as a DAI biomarker. PRPH, as a 57 kDa type III intermediate filament protein, forms a part of the cytoskeleton in a subset of neurons (of the peripheral nervous system and central nervous system) that have peripheral axons (Hol et al., 2017; Portier et al., 1984). Early studies were suggestive of participation of PRPH in axon outgrowth and regeneration during the development of the nervous system (Escurat M et al., 1990; Troy et al., 1990a). Additional studies have implicated the interactions of PRPH with NF proteins and showed that PRPH could coassemble with each NF subunit and that both PRPH and NFs are deeply involved in amyotrophic lateral sclerosis (Beaulieu et al., 1999; Parysek et al., 1991; Xiao et al., 2006). Moreover, NFs and PRPH manifest the opposite temporal patterns of gene expression in response to injury: In contrast to the NFs, PRPH is strongly upregulated in motor neurons after traumatic neuronal injury (Chadan et al., 1994; Oblinger et al., 1989; Troy et al., 1990b; Wong et al., 1990). Additionally, PRPH expression is induced after brain injuries in subsets of neurons where it is usually silent, and even in varicose swelling axons (Beaulieu et al., 2002). These data suggest that PRPH may perform a notable function in neuronal regeneration. Similarly, in our iTRAQ data, PRPH amounts significantly increased with the downregulation of NF proteins, thus pointing to a different function of PRPH in the development and progression of DAI-related pathologies as compared to the NFs. Furthermore, PRPH manifested significantly large fold changes at all the time points postinjury, as successfully confirmed by

western blotting. Meanwhile, IHC staining of PRPH revealed that its profiles were mainly found in axons, and the number of PRPH positive axons with the morphological appearance of injured axons was increased in the corpus callosum following DAI. Taken together, these findings indicate its good potential as a DAI biomarker.

4.3 Activation of caspase-mediated programmed cell death during DAI

The Ca^{2+} overload with the activation of calpains during DAI also leads to mitochondrial dysfunction, which causes disruption of energy metabolism and ion homeostasis, with a subsequent release of cytochrome c and activation of caspases (Buki et al., 2000; Buki and Povlishock, 2006). Caspases are a family of cysteine proteases that perform an important function in apoptosis after TBI, especially caspase 3 (Clark et al., 2000; Yuan et al., 2000). Thus, caspase-mediated neuronal apoptosis is a major contributor to the secondary neuronal injury during DAI. Here, we found 58 DEPs, including Kcnip3, Prkcd, Alb and Anp32a, related to the apoptotic process (Table 3). Kcnip3, as noted above, is known to be a proapoptotic factor, meaning that it has a dual role in apoptosis and Ca^{2+} modulation in response to DAI. Protein kinase C δ type (Prkcd), a novel member of the PKC family, participates in the induction of apoptosis (Brodie et al., 2003). Prkcd can be activated by many extracellular factors, including excitotoxic injury (Kaasinen et al., 2002), pro-inflammatory cytokines (Wallerstedt et al., 2010), and oxidative stress (Qi et al., 2011), all of which are known to be associated with secondary injury after TBI (Park et al., 2008). Both Kcnip3 and Prkcd were found to be significantly upregulated postinjury in this study, indicating that they may be crucial for the apoptotic process of DAI in rats.

4.4 The inflammatory response during DAI

Increasing evidence suggests that neuroinflammatory processes are a major contributor to the pathophysiology of the secondary neural injury during DAI (Csuka et al., 2000; Ekmark-Lewén S et al., 2013; Lin and Wen, 2013; Pang et al., 2016). The complement system, which is a powerful pillar of the innate immune system, has been

suggested to participate in the acute inflammatory pathophysiology of TBI by mediating cerebral edema and delayed neuronal cell death (Bellander et al., 2001). Elevated postinjury C3 levels were found in the cerebrospinal fluid of patients with severe TBI (Kossmann et al., 1997). Increased immunoreactivity of C1q, C3, C3b, C3d, and C5b-9 is found on neurons in the penumbra region of the contused brain area (Bellander et al., 2001). In a study on an animal TBI model, immunoreactivity for C3 was found around the lesion but not in the uninjured contralateral hemisphere (Bellander et al., 1996). Collectively, these findings indicate that complement pathways are activated in response to TBI, thereby contributing to nerve cell death. One mechanism underlying the elevated complement levels after TBI is induction of the acute phase response after trauma; this phenomenon enhances the synthesis of acute phase proteins, such as C-reactive protein (Suankratay et al., 1998; Szalai et al., 2000). In our iTRAQ data, increased expression of C3 was noted 1 and 3 h postinjury, pointing to complement activation in the early stage of DAI. Furthermore, significant upregulation of C-reactive protein was found in our study. As a result, we can speculate that neuroinflammatory processes, especially activation of the complement system, participate in secondary neuronal cell death during DAI, probably via induction of the acute phase response.

4.5 Limitations and outlooks

The fact that only the time point 72 h has been controlled using sham rats is a major limitation of the study, due to the fact that it may have lead to false positives in the groups of rats with earlier survival time points. The main considerations are as follows: 1) The sham rats underwent surgical procedure of scalp incision without the impact, the damage to our samples (brain tissues) was minor to the impact injury of the rats in DAI group. According to our observations, the symptoms of the rats after DAI were mainly caused by brain injury, especially those with earlier survival time points (within 48 h postinjury) which had more severe symptoms. 2) Due to the minute quantity of corpus callosum, ten samples were pooled within each group to ensure sufficient amounts of tissue samples for iTRAQ analysis, in each group. Thus,

in order to ensure the uniformity of the sham group, rats executed at the same time (72 h) were selected.

On the other hand, the following aspects need further study: 1) Double immunofluorescence staining will be performed to further determine the cell types in c in the brain following DAI. 2) Since both PRPH and CSEN were observed in axons, double immunofluorescence staining with β -APP will be performed to evaluate for their potential co-localization and further clarify their potential as biomarkers of damaged axons. 3) Future study of these DEPs will be extended to human brain tissues to further examine the mechanisms of axonal damage in human.

Conclusion

In this study, we performed iTRAQ to determine protein expression patterns via unbiased quantitative proteomic profiling in order to investigate changes in the expression of corpus callosum proteins during DAI in rats. A total of 4460 proteins were detected, 514 of which were differentially expressed between the injury groups and the control. Bioinformatic analysis revealed some enriched GO and KEGG pathway terms in this set of proteins. In addition, protein–protein interaction analysis uncovered six clusters of the interaction networks. On the basis of the bioinformatic analysis of these DEPs, we can speculate about the pathogenesis of DAI, which appears to involve calcium ion–regulatory proteins (e.g., CSEN and ryanodine receptor 2), cytoskeleton organization (e.g., NFL, NFM, NFH, and PRPH), an apoptotic process (e.g., CSEN, Prkcd, and Anp32a), and inflammatory-response proteins (e.g., C3 and C-reactive protein). Notably, PRPH and CSEN—which are thought to be associated with cytoskeleton alterations, calcium ion regulation, and an apoptotic process—were significantly upregulated during DAI, indicating their good potential as DAI biomarkers. This work seems to provide a detailed understanding of the mechanisms of axonal damage in rats during DAI.

Conflict of interest

All authors declare that they have no conflicts of interest to this work.

Compliance with ethical standards

The experiments in this study were conducted in accordance with the local guidelines for the care and use of animals.

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References:

- An, W.F., Bowlby, M.R., Betty, M., Cao, J., Ling, H.P., Mendoza, G., Hinson, J.W., Mattsson, K.I., Strassle, B.W., Trimmer, J.S., Rhodes, K.J., 2000. Modulation of A-type potassium channels by a family of calcium sensors. *Nature*. 403 (6769), 553-6.
- Beaulieu, J.M., Robertson, J., Julien, J.P., 1999. Interactions between peripherin and neurofilaments in cultured cells: disruption of peripherin assembly by the NF-M and NF-H subunits. *Biochem Cell Biol*. 77 (1), 41-5.
- Beaulieu, J.M., Kriz, J., Julien, J.P., 2002. Induction of peripherin expression in subsets of brain neurons after lesion injury or cerebral ischemia. *Brain Res*. 946 (2), 153-61.
- Bellander, B.M., von Holst, H., Fredman, P., Svensson, M., 1996. Activation of the complement cascade and increase of clusterin in the brain following a cortical contusion in the adult rat. *J Neurosurg*. 85 (3), 468-75.
- Bellander, B.M., Singhrao, S.K., Ohlsson, M., Mattsson, P., Svensson, M., 2001. Complement activation in the human brain after traumatic head injury. *J Neurotrauma*. 18 (12), 1295-311.
- Brodie, C., Blumberg, P.M., 2003. Regulation of cell apoptosis by protein kinase c delta. *Apoptosis*. 8 (1), 19-27.
- Buki, A., Okonkwo, D.O., Wang, K.K., Povlishock, J.T., 2000. Cytochrome c release and caspase activation in traumatic axonal injury. *J Neurosci*. 20 (8), 2825-34.
- Buki, A., Povlishock, J.T., 2006. All roads lead to disconnection?--Traumatic axonal injury revisited. *Acta Neurochir (Wien)*. 148 (2), 181-93; discussion 193-4.
- Buxbaum, J.D., Choi, E.K., Luo, Y., Lilliehook, C., Crowley, A.C., Merriam, D.E., Wasco, W., 1998. Calsenilin: a calcium-binding protein that interacts with the presenilins and regulates the levels of a presenilin fragment. *Nat Med*. 4 (10), 1177-81.
- Carrion, A.M., Link, W.A., Ledo, F., Mellstrom, B., Naranjo, J.R., 1999. DREAM is a Ca²⁺-regulated transcriptional repressor. *Nature*. 398 (6722), 80-4.
- Cengiz, P., Zemlan, F., Ellenbogen, R., Hawkins, D., Zimmerman, J.J., 2008. Cerebrospinal fluid cleaved-tau protein and 9-hydroxyoctadecadienoic acid concentrations in pediatric patients with hydrocephalus. *Pediatr Crit Care Med*. 9 (5), 524-9.
- Chadan, S., Le Gall, J.Y., Di Giamberardino, L., Filliatreau, G., 1994. Axonal transport of type III intermediate filament protein peripherin in intact and regenerating motor axons of the rat sciatic nerve. *J Neurosci Res*. 39 (2), 127-39.
- Chang, M.C., Jang, S.H., 2010. Corpus callosum injury in patients with diffuse axonal injury: a diffusion tensor imaging study. *NeuroRehabilitation*. 26 (4), 339-45.
- Choi, E.K., Zaidi, N.F., Miller, J.S., Crowley, A.C., Merriam, D.E., Lilliehook, C., Buxbaum, J.D., Wasco, W., 2001. Calsenilin is a substrate for caspase-3 that preferentially interacts with the familial Alzheimer's disease-associated C-terminal fragment of presenilin 2. *J Biol Chem*. 276 (22), 19197-204.
- Cicuendez, M., Castano-Leon, A., Ramos, A., Hilario, A., Gomez, P.A., Lagares, A., 2017. Prognostic value of corpus callosum injuries in severe head trauma. *Acta Neurochir (Wien)*. 159 (1), 25-32.
- Clark, R.S., Kochanek, P.M., Watkins, S.C., Chen, M., Dixon, C.E., Seidberg, N.A., Melick, J., Loeffert, J.E.,

- Nathaniel, P.D., Jin, K.L., Graham, S.H., 2000. Caspase-3 mediated neuronal death after traumatic brain injury in rats. *J Neurochem.* 74 (2), 740-53.
- Crawford, F., Crynen, G., Reed, J., Mouzon, B., Bishop, A., Katz, B., Ferguson, S., Phillips, J., Ganapathi, V., Mathura, V., Roses, A., Mullan, M., 2012. Identification of plasma biomarkers of TBI outcome using proteomic approaches in an APOE mouse model. *J Neurotrauma.* 29 (2), 246-60.
- Csuka, E., Hans, V.H., Ammann, E., Trentz, O., Kossmann, T., Morganti-Kossmann, M.C., 2000. Cell activation and inflammatory response following traumatic axonal injury in the rat. *Neuroreport.* 11 (11), 2587-90.
- Czogalla, A., Sikorski, A.F., 2005. Spectrin and calpain: a 'target' and a 'sniper' in the pathology of neuronal cells. *Cell Mol Life Sci.* 62 (17), 1913-24.
- Datta, A., Akatsu, H., Heese, K., Sze, S.K., 2013. Quantitative clinical proteomic study of autopsied human infarcted brain specimens to elucidate the deregulated pathways in ischemic stroke pathology. *J Proteomics.* 91, 556-68.
- Dolinak, D., Smith, C., Graham, D.I., 2000. Hypoglycaemia is a cause of axonal injury. *Neuropathol Appl Neurobiol.* 26 (5), 448-53.
- Ekmark-Lewén S, Flygt J, Kiwanuka O, Meyerson BJ, Lewén A, Hillered L, N, M., 2013. Traumatic axonal injury in the mouse is accompanied by a dynamic inflammatory response, astroglial reactivity and complex behavioral changes. *J Neuroinflammation.* 10, 44.
- Escurat M, Djabali K, Gumpel M, Gros F, MM, P., 1990. Differential expression of two neuronal intermediate-filament proteins, peripherin and the low-molecular-mass neurofilament protein (NF-L), during the development of the rat. *J Neurosci.* 10 (3), 764-84.
- Ezaki, Y., Tsutsumi, K., Morikawa, M., Nagata, I., 2006. Role of diffusion-weighted magnetic resonance imaging in diffuse axonal injury. *Acta Radiol.* 47 (7), 733-40.
- Foda, M.A., Marmarou, A., 1994. A new model of diffuse brain injury in rats. Part II: Morphological characterization. *J Neurosurg.* 80 (2), 301-13.
- Frati, A., Cerretani, D., Fiaschi, A.I., Frati, P., Gatto, V., La Russa, R., Pesce, A., Pinchi, E., Santurro, A., Frascchetti, F., Fineschi, V., 2017. Diffuse Axonal Injury and Oxidative Stress: A Comprehensive Review. *Int J Mol Sci.* 18 (12).
- Goodman, S.R., Zimmer, W.E., Clark, M.B., Zagon, I.S., Barker, J.E., Bloom, M.L., 1995. Brain spectrin: of mice and men. *Brain Res Bull.* 36 (6), 593-606.
- Hammond, P.I., Craig, T.A., Kumar, R., Brimijoin, S., 2003. Regional and cellular distribution of DREAM in adult rat brain consistent with multiple sensory processing roles. *Molecular Brain Research.* 111 (1-2), 104-110.
- Hol, E.M., Capetanaki, Y., 2017. Type III Intermediate Filaments Desmin, Glial Fibrillary Acidic Protein (GFAP), Vimentin, and Peripherin. *Cold Spring Harb Perspect Biol.* 9 (12), pii: a021642.
- Hortobagyi, T., Wise, S., Hunt, N., Cary, N., Djurovic, V., Fegan-Earl, A., Shorrock, K., Rouse, D., Al-Sarraj, S., 2007. Traumatic axonal damage in the brain can be detected using beta-APP immunohistochemistry within 35 min after head injury to human adults. *Neuropathol Appl Neurobiol.* 33 (2), 226-37.
- Hulkower, M.B., Poliak, D.B., Rosenbaum, S.B., Zimmerman, M.E., Lipton, M.L., 2013. A decade of DTI in traumatic brain injury: 10 years and 100 articles later. *AJNR Am J Neuroradiol.* 34 (11), 2064-74.
- Jensen, L.J., Kuhn, M., Stark, M., Chaffron, S., Creevey, C., Muller, J., Doerks, T., Julien, P., Roth, A.,

- Simonovic, M., Bork, P., von Mering, C., 2009. STRING 8--a global view on proteins and their functional interactions in 630 organisms. *Nucleic Acids Res.* 37 (Database issue), D412-6.
- Jin, J.K., Choi, J.K., Wasco, W., Buxbaum, J.D., Kozłowski, P.B., Carp, R.I., Kim, Y.S., Choi, E.K., 2005. Expression of calsenilin in neurons and astrocytes in the Alzheimer's disease brain. *Neuroreport.* 16 (5), 451-5.
- Jo, D.G., Jang, J., Kim, B.J., Lundkvist, J., Jung, Y.K., 2005. Overexpression of calsenilin enhances gamma-secretase activity. *Neurosci Lett.* 378 (1), 59-64.
- Johnson, V.E., Stewart, W., Smith, D.H., 2013. Axonal pathology in traumatic brain injury. *Exp Neurol.* 246, 35-43.
- Johnson, V.E., Stewart, W., Weber, M.T., Cullen, D.K., Siman, R., Smith, D.H., 2016. SNTF immunostaining reveals previously undetected axonal pathology in traumatic brain injury. *Acta Neuropathol.* 131 (1), 115-35.
- Kaasinen, S.K., Goldsteins, G., Alhonen, L., Jänne, J., Koistinaho, J., 2002. Induction and Activation of Protein Kinase C δ in Hippocampus and Cortex after Kainic Acid Treatment. *Experimental Neurology.* 176 (1), 203-212.
- Kaur, B., Ratty, G.N., Timperley, W.R., 1999. The possible role of hypoxia in the formation of axonal bulbs. *J Clin Pathol.* 52 (3), 203-9.
- Kobek, M., Skowronek, R., Jankowski, Z., Pałasz, A., 2014. Neurofilaments and traumatic brain injury. *Archives of Forensic Medicine and Criminology.* 64 (4), 268-279.
- Kossmann, T., Stahel, P.F., Morganti-Kossmann, M.C., Jones, J.L., Barnum, S.R., 1997. Elevated levels of the complement components C3 and factor B in ventricular cerebrospinal fluid of patients with traumatic brain injury. *J Neuroimmunol.* 73 (1-2), 63-9.
- Kupina, N.C., Detloff, M.R., Dutta, S., Hall, E.D., 2002. Neuroimmunophilin ligand V-10,367 is neuroprotective after 24-hour delayed administration in a mouse model of diffuse traumatic brain injury. *J Cereb Blood Flow Metab.* 22 (10), 1212-21.
- Lam, H., Deutsch, E.W., Aebersold, R., 2010. Artificial decoy spectral libraries for false discovery rate estimation in spectral library searching in proteomics. *J Proteome Res.* 9 (1), 605-10.
- Larrode, P., Calvo, A.C., Moreno-Martinez, L., de la Torre, M., Moreno-Garcia, L., Molina, N., Castiella, T., Iniguez, C., Pascual, L.F., Mena, F.J.M., Zaragoza, P., SR, Y.C., Osta, R., 2018. DREAM-Dependent Activation of Astrocytes in Amyotrophic Lateral Sclerosis. *Mol Neurobiol.* 55 (1), 1-12.
- Lepinoux-Chambaud, C., Eyer, J., 2013. Review on intermediate filaments of the nervous system and their pathological alterations. *Histochem Cell Biol.* 140 (1), 13-22.
- Lilliehook, C., Chan, S., Choi, E.K., Zaidi, N.F., Wasco, W., Mattson, M.P., Buxbaum, J.D., 2002. Calsenilin enhances apoptosis by altering endoplasmic reticulum calcium signaling. *Mol Cell Neurosci.* 19 (4), 552-9.
- Lin, Y., Wen, L., 2013. Inflammatory response following diffuse axonal injury. *Int J Med Sci.* 10 (5), 515-21.
- Liu, J., Liu, M.C., Wang, K.K., 2008. Calpain in the CNS: from synaptic function to neurotoxicity. *Sci Signal.* 1 (14), re1.
- Ljungqvist, J., Zetterberg, H., Mitsis, M., Blennow, K., Skoglund, T., 2017. Serum Neurofilament Light Protein as a Marker for Diffuse Axonal Injury: Results from a Case Series Study. *J Neurotrauma.* 34 (5), 1124-1127.
- Ma, J., Zhang, K., Wang, Z., Chen, G., 2016. Progress of Research on Diffuse Axonal Injury after

- Traumatic Brain Injury. *Neural Plast.* 2016, 9746313.
- Manivannan, S., Makwana, M., Ahmed, A.I., Zaben, M., 2018. Profiling biomarkers of traumatic axonal injury: From mouse to man. *Clin Neurol Neurosurg.* 171, 6-20.
- Marmarou, A., Foda, M.A., van den Brink, W., Campbell, J., Kita, H., Demetriadou, K., 1994. A new model of diffuse brain injury in rats. Part I: Pathophysiology and biomechanics. *J Neurosurg.* 80 (2), 291-300.
- McCracken, E., Hunter, A.J., Patel, S., Graham, D.I., Dewar, D., 1999. Calpain activation and cytoskeletal protein breakdown in the corpus callosum of head-injured patients. *J Neurotrauma.* 16 (9), 749-61.
- McKenzie, K.J., McLellan Dr Fau - Gentleman, S.M., Gentleman Sm Fau - Maxwell, W.L., Maxwell WI Fau - Gennarelli, T.A., Gennarelli Ta Fau - Graham, D.I., Graham, D.I., 1996. Is beta-APP a marker of axonal damage in short-surviving head injury? *Acta Neuropathol.* 92 (6), 608-13.
- Meythaler, J.M., Peduzzi, J.D., Eleftheriou, E., Novack, T.A., 2001. Current concepts: Diffuse axonal injury—associated traumatic brain injury. *Archives of Physical Medicine and Rehabilitation.* 82 (10), 1461-1471.
- Muth, T., Keller, D., Puetz, S.M., Martens, L., Sickmann, A., Boehm, A.M., 2010. jTraqX: a free, platform independent tool for isobaric tag quantitation at the protein level. *Proteomics.* 10 (6), 1223-5.
- Oblinger, M.M., Wong, J., Parysek, L.M., 1989. Axotomy-induced changes in the expression of a type III neuronal intermediate filament gene. *J Neurosci.* 9 (11), 3766-75.
- Oehmichen, M., Meissner, C., Schmidt, V., Pedal, I., Konig, H.G., 1999. Pontine axonal injury after brain trauma and nontraumatic hypoxic-ischemic brain damage. *Int J Legal Med.* 112 (4), 261-7.
- Pang, H., Huang, T., Song, J., Li, D., Zhao, Y., Ma, X., 2016. Inhibiting HMGB1 with Glycyrrhizic Acid Protects Brain Injury after DAI via Its Anti-Inflammatory Effect. *Mediators Inflamm.* 2016, 4569521.
- Park, E., Bell, J.D., Baker, A.J., 2008. Traumatic brain injury: can the consequences be stopped? *CMAJ.* 178 (9), 1163-70.
- Parysek, L.M., McReynolds, M.A., Goldman, R.D., Ley, C.A., 1991. Some neural intermediate filaments contain both peripherin and the neurofilament proteins. *J Neurosci Res.* 30 (1), 80-91.
- Paxinos, G., Watson, C., 2006. The rat brain in stereotaxic coordinates, Vol. 3, Academic Press, San Diego, CA.
- Perrot, R., Berges, R., Bocquet, A., Eyer, J., 2008. Review of the multiple aspects of neurofilament functions, and their possible contribution to neurodegeneration. *Mol Neurobiol.* 38 (1), 27-65.
- Portier, M.M., de Nechaud, B., Gros, F., 1984. Peripherin, a new member of the intermediate filament protein family. *Dev Neurosci.* 6 (6), 335-344.
- Posmantur, R., Hayes, R.L., Dixon, C.E., Taft, W.C., 1994. Neurofilament 68 and neurofilament 200 protein levels decrease after traumatic brain injury. *J Neurotrauma.* 11 (5), 533-45.
- Qi, X., Disatnik, M.H., Shen, N., Sobel, R.A., Mochly-Rosen, D., 2011. Aberrant mitochondrial fission in neurons induced by protein kinase C δ under oxidative stress conditions in vivo. *Mol Biol Cell.* 22 (2), 256-65.
- Reeves, T.M., Greer, J.E., Vanderveer, A.S., Phillips, L.L., 2010. Proteolysis of submembrane cytoskeletal proteins ankyrin-G and α II-spectrin following diffuse brain injury: a role in white matter vulnerability at Nodes of Ranvier. *Brain Pathol.* 20 (6), 1055-68.
- Ren, T., Lin, S., Wang, Z., Shang, A., 2016. Differential proteomics analysis of low- and high-grade of

- astrocytoma using iTRAQ quantification. *Onco Targets Ther.* 9, 5883-5895.
- Ross, P.L., Huang, Y.N., Marchese, J.N., Williamson, B., Parker, K., Hattan, S., Khainovski, N., Pillai, S., Dey, S., Daniels, S., Purkayastha, S., Juhasz, P., Martin, S., Bartlett-Jones, M., He, F., Jacobson, A., Pappin, D.J., 2004. Multiplexed protein quantitation in *Saccharomyces cerevisiae* using amine-reactive isobaric tagging reagents. *Mol Cell Proteomics.* 3 (12), 1154-69.
- Schwacke, J.H., Hill, E.G., Krug, E.L., Comte-Walters, S., Schey, K.L., 2009. iQuantitor: a tool for protein expression inference using iTRAQ. *BMC Bioinformatics.* 10, 342.
- Sherriff, F.E., Bridges, L.R., Sivaloganathan, S., 1994. Early detection of axonal injury after human head trauma using immunocytochemistry for beta-amyloid precursor protein. *Acta Neuropathol.* 87 (1), 55-62.
- Shilov, I.V., Seymour, S.L., Patel, A.A., Loboda, A., Tang, W.H., Keating, S.P., Hunter, C.L., Nuwaysir, L.M., Schaeffer, D.A., 2007. The Paragon Algorithm, a next generation search engine that uses sequence temperature values and feature probabilities to identify peptides from tandem mass spectra. *Mol Cell Proteomics.* 6 (9), 1638-55.
- Siedler, D.G., Chuah, M.I., Kirkcaldie, M.T., Vickers, J.C., King, A.E., 2014. Diffuse axonal injury in brain trauma: insights from alterations in neurofilaments. *Front Cell Neurosci.* 8, 429.
- Smith, D.H., Meaney, D.F., 2000. Axonal damage in traumatic brain injury. *Neuroscientist.* 6 (6), 483-495.
- Smith, D.H., Meaney, D.F., Shull, W.H., 2003. Diffuse axonal injury in head trauma. *J Head Trauma Rehabil.* 18 (4), 307-16.
- Staal, J.A., Dickson, T.C., Gasperini, R., Liu, Y., Foa, L., Vickers, J.C., 2010. Initial calcium release from intracellular stores followed by calcium dysregulation is linked to secondary axotomy following transient axonal stretch injury. *J Neurochem.* 112 (5), 1147-55.
- Suankratay, C., Mold, C., Zhang, Y., Potempa, L.A., Lint, T.F., Gewurz, H., 1998. Complement regulation in innate immunity and the acute-phase response: inhibition of mannan-binding lectin-initiated complement cytolysis by C-reactive protein (CRP). *Clin Exp Immunol.* 113 (3), 353-9.
- Szalai, A.J., van Ginkel, F.W., Wang, Y., McGhee, J.R., Volanakis, J.E., 2000. Complement-dependent acute-phase expression of C-reactive protein and serum amyloid P-component. *J Immunol.* 165 (2), 1030-5.
- Tang-Schomer, M.D., Johnson, V.E., Baas, P.W., Stewart, W., Smith, D.H., 2012. Partial interruption of axonal transport due to microtubule breakage accounts for the formation of periodic varicosities after traumatic axonal injury. *Exp Neurol.* 233 (1), 364-72.
- Troy, C.M., Brown, K., Greene, L.A., Shelanski, M.L., 1990a. Ontogeny of the neuronal intermediate filament protein, peripherin, in the mouse embryo. *Neuroscience.* 36 (1), 217-37.
- Troy, C.M., Muma, N.A., Greene, L.A., Price, D.L., Shelanski, M.L., 1990b. Regulation of peripherin and neurofilament expression in regenerating rat motor neurons. *Brain Res.* 529 (1-2), 232-8.
- Tsitsopoulos, P.P., Abu Hamdeh, S., Marklund, N., 2017. Current Opportunities for Clinical Monitoring of Axonal Pathology in Traumatic Brain Injury. *Front Neurol.* 8, 599.
- Ubukata, S., Ueda, K., Sugihara, G., Yassin, W., Aso, T., Fukuyama, H., Murai, T., 2016. Corpus Callosum Pathology as a Potential Surrogate Marker of Cognitive Impairment in Diffuse Axonal Injury. *J Neuropsychiatry Clin Neurosci.* 28 (2), 97-103.
- Valiyaveetil, M., Alamneh, Y.A., Wang, Y., Arun, P., Oguntayo, S., Wei, Y., Long, J.B., Nambiar, M.P., 2014. Cytoskeletal protein alpha-II spectrin degradation in the brain of repeated blast exposed mice. *Brain Res.* 1549, 32-41.

- Wallerstedt, E., Smith, U., Andersson, C.X., 2010. Protein kinase C-delta is involved in the inflammatory effect of IL-6 in mouse adipose cells. *Diabetologia*. 53 (5), 946-54.
- Winkler, T., Sharma, H.S., Stalberg, E., Badgaiyan, R.D., Gordh, T., Westman, J., 2003. An L-type calcium channel blocker, nimodipine influences trauma induced spinal cord conduction and axonal injury in the rat. *Acta Neurochir Suppl*. 86, 425-32.
- Wong, J., Oblinger, M.M., 1990. Differential regulation of peripherin and neurofilament gene expression in regenerating rat DRG neurons. *J Neurosci Res*. 27 (3), 332-41.
- Xiao, S., McLean, J., Robertson, J., 2006. Neuronal intermediate filaments and ALS: a new look at an old question. *Biochim Biophys Acta*. 1762 (11-12), 1001-12.
- Yuan, J., Yankner, B.A., 2000. Apoptosis in the nervous system. *Nature*. 407 (6805), 802-9.

Figure legends:

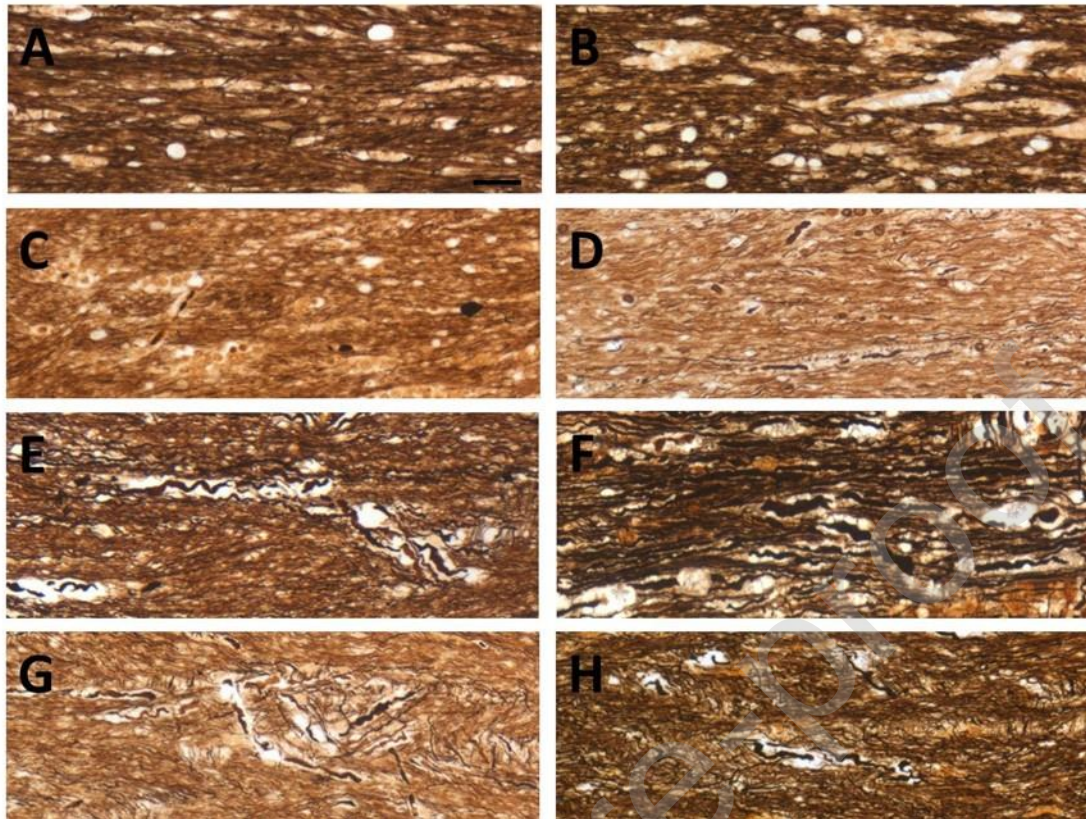


Fig. 1 Silver staining in the corpus callosum of sham and DAI rats (bar = 50 μ m). In DAI group, the axons showed various degrees of swollen, twisted, rupture and ARBs formation as early as 3h postinjury (B-H), with normal axons noted in the sham group (A).

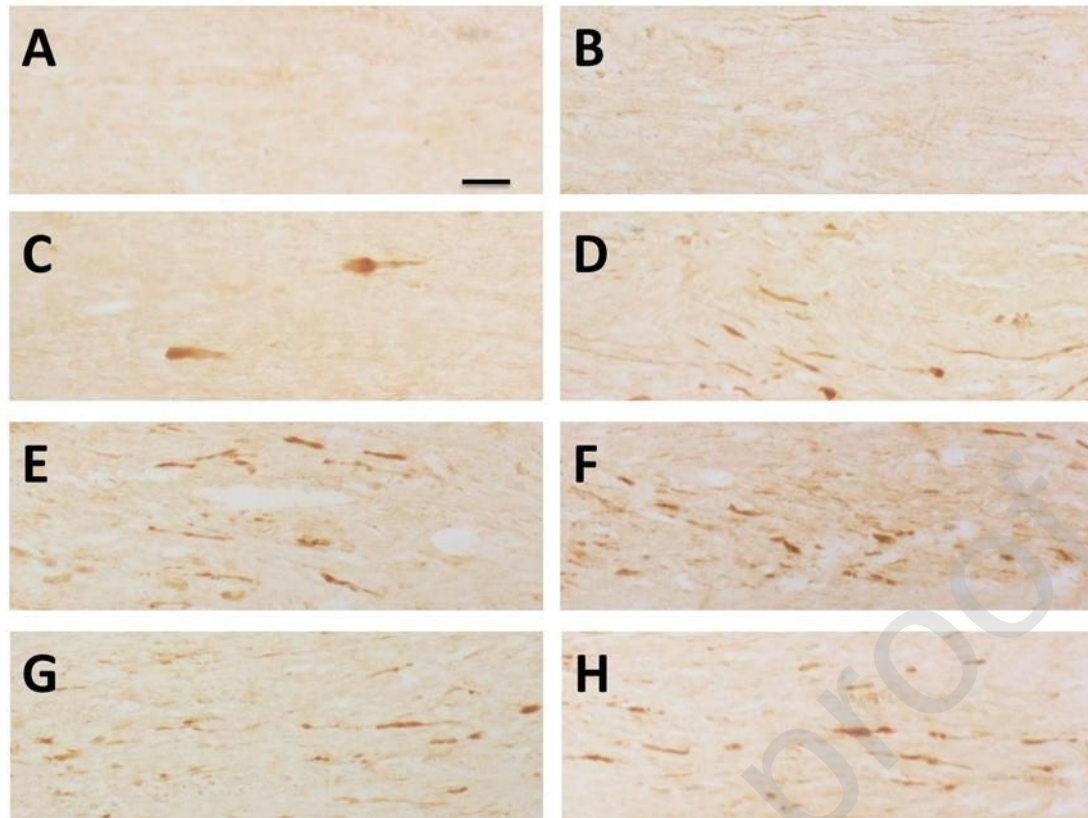


Fig. 2. β -APP immunohistochemistry staining in the corpus callosum of sham and DAI rats (A. sham operated; B. 1 h postinjury; C. 3 h postinjury; D. 6 h postinjury; E. 12 h postinjury; F. 24 h postinjury; G. 48 h postinjury; H. 72 h postinjury, bar = 50 μ m).

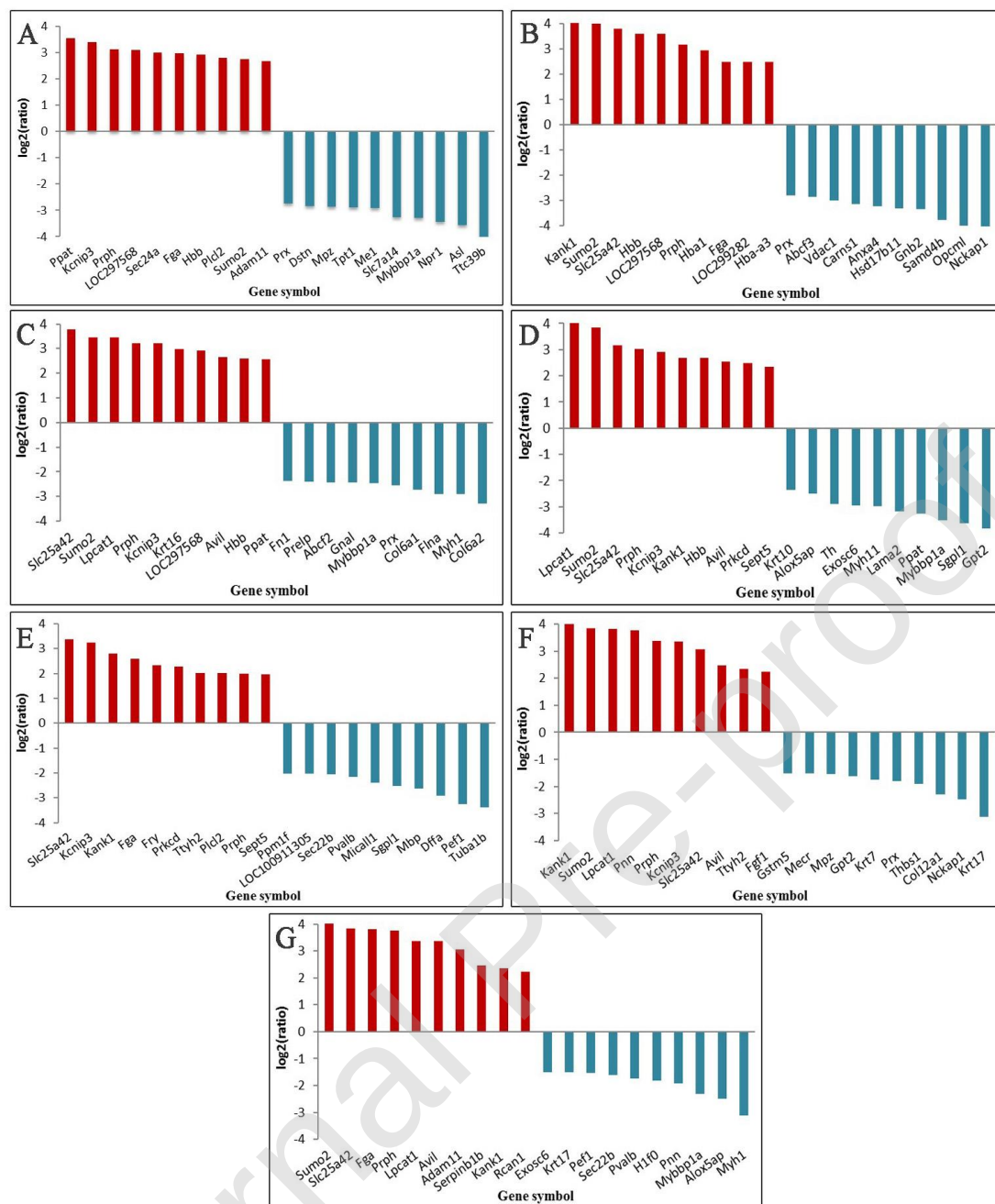


Fig. 3 The top 10 up-regulated and down-regulated differential proteins and their fold changes at all seven time points postinjury (A. 1 h postinjury; B. 3 h postinjury; C. 6 h postinjury; D. 12 h postinjury; E. 24 h postinjury; F. 48 h postinjury; G. 72 h postinjury).

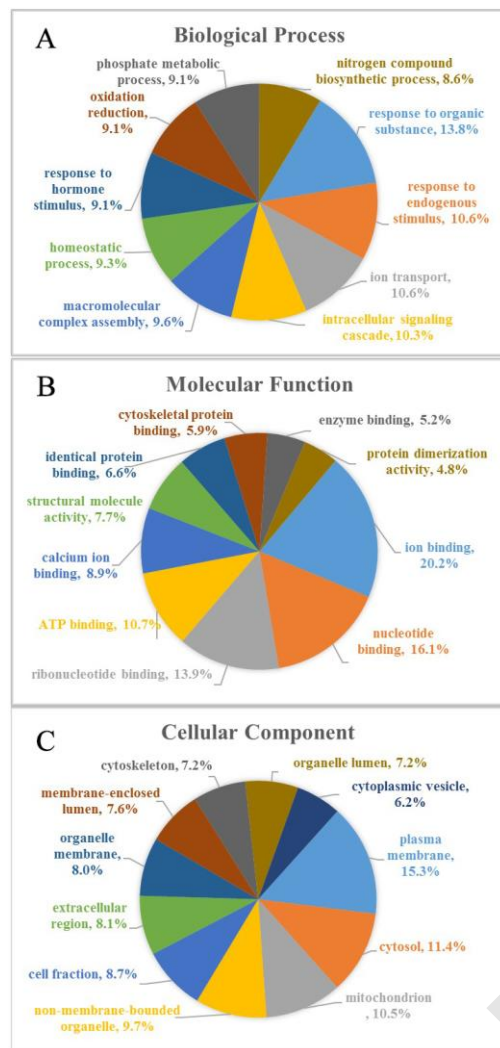


Fig. 4 Gene ontology (GO) analysis of 514 differentially expressed proteins identified in this study involved in biological process (A), molecular function (B) and cellular component (C) against the GO database using DAVID analysis.

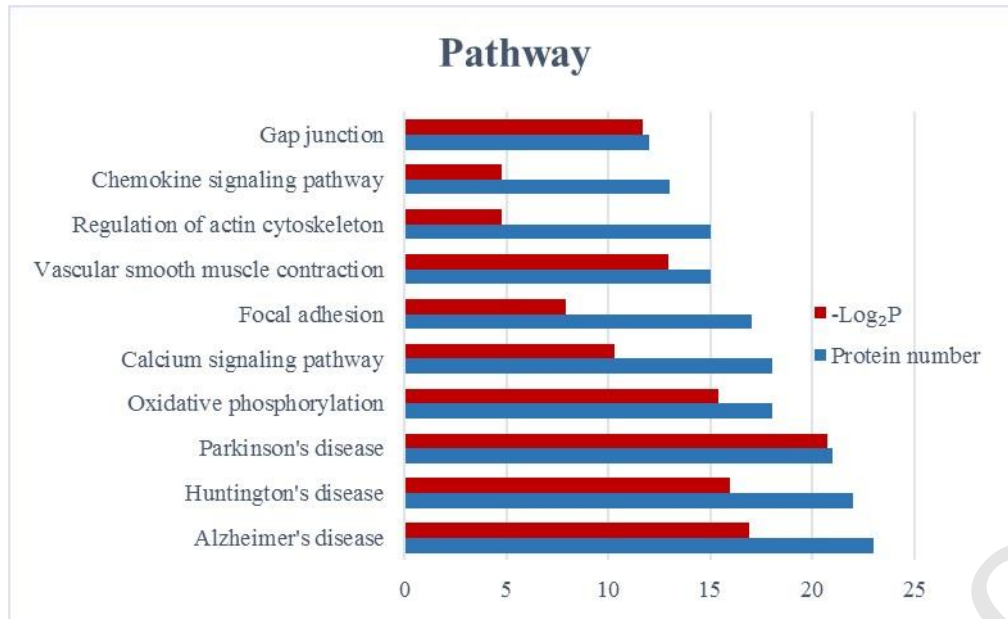


Fig. 5 The top 10 enriched KEGG pathways of differential proteins.

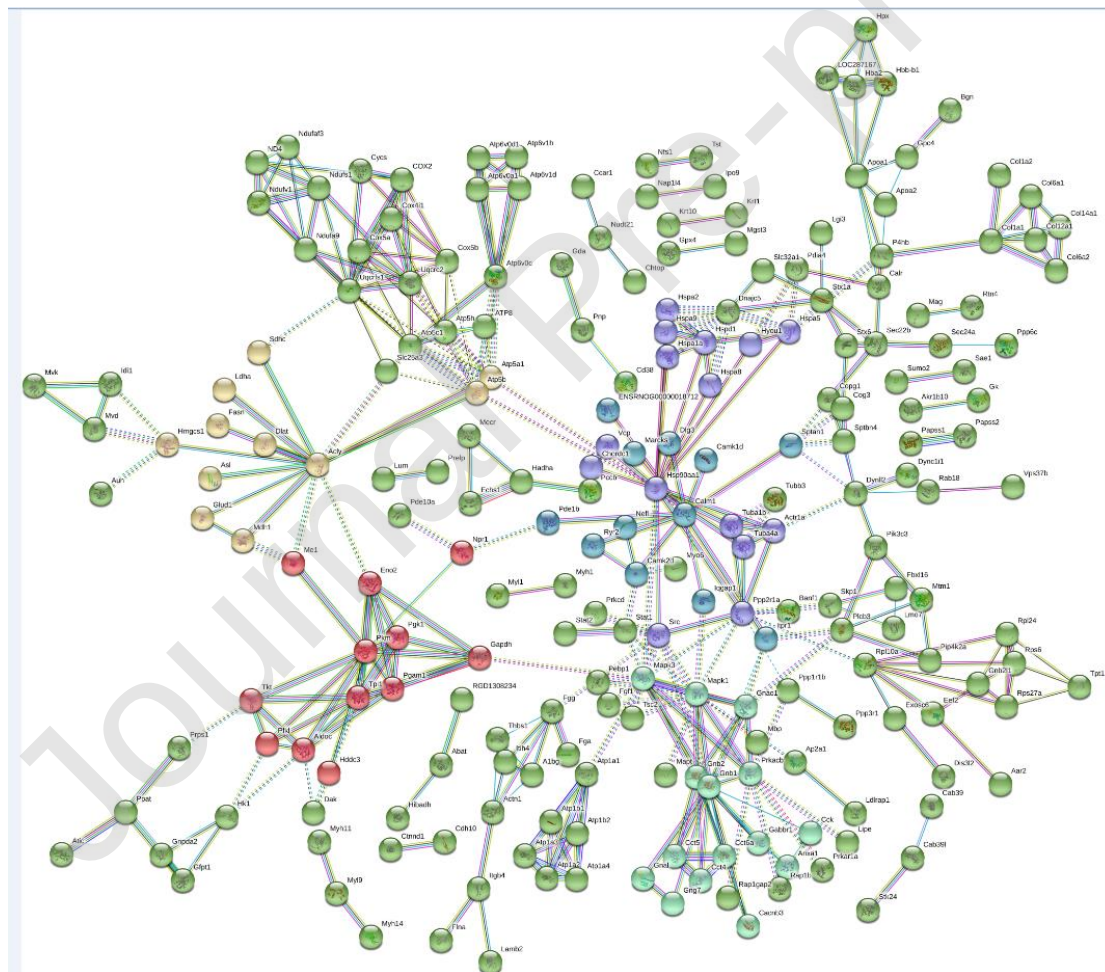


Fig. 6 Interaction network analysis of 514 differential proteins analyzed by STRING. The minimum required interaction score was set as the highest confidence (0.900) and the disconnected nodes in the network were hidden. The network map was clustered

in 6 clusters using K-means method. In this network, nodes denote proteins. Colored lines between the proteins represent the types of interaction evidence (sky-blue and purple lines: known interactions; green, red and blue lines: predicted interactions; other colored lines: textmining, co-expression and protein homology).

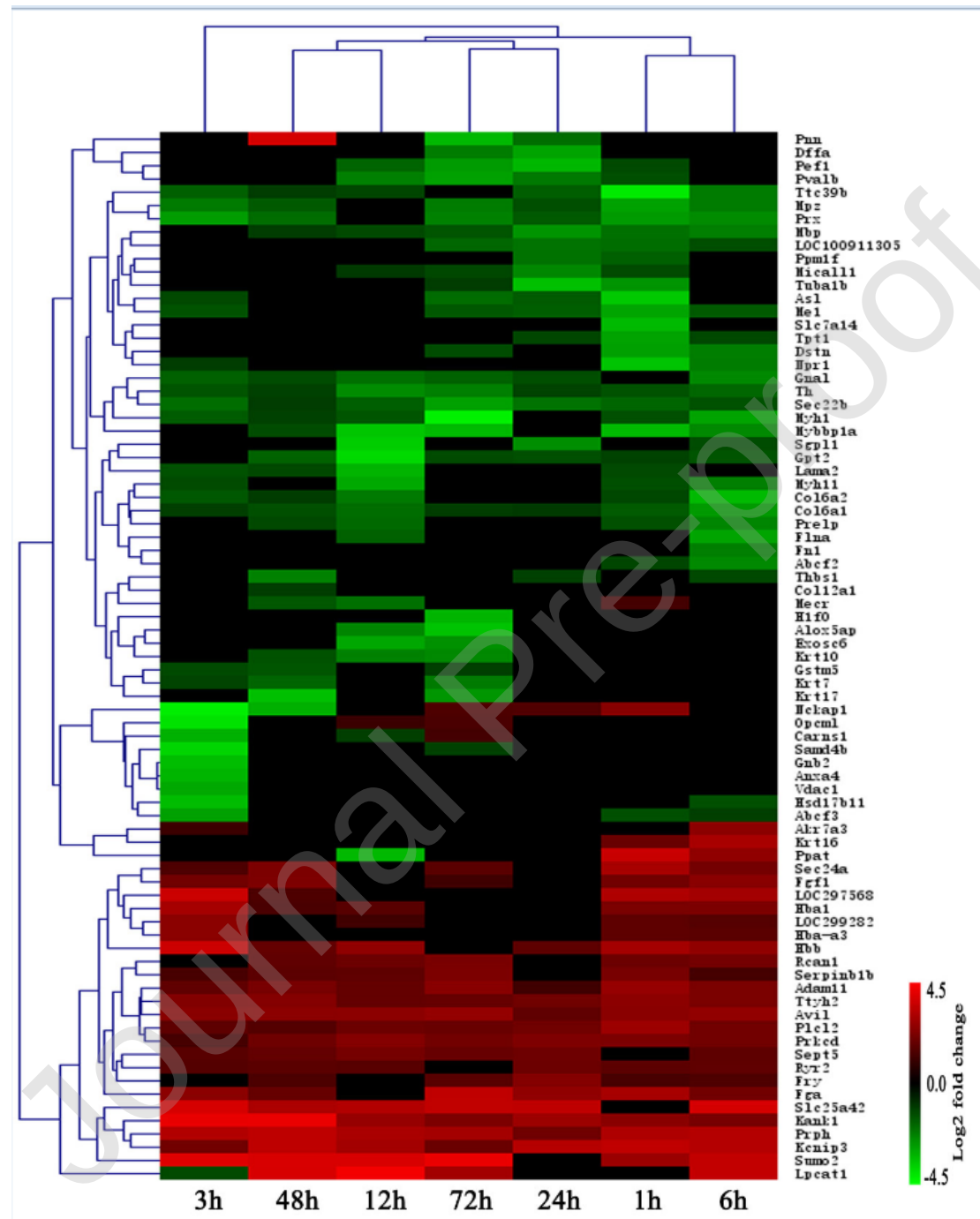


Fig. 7 Hierarchical clustering analysis of the top ten up-regulated and down-regulated differential proteins at each time point of the DAI groups.

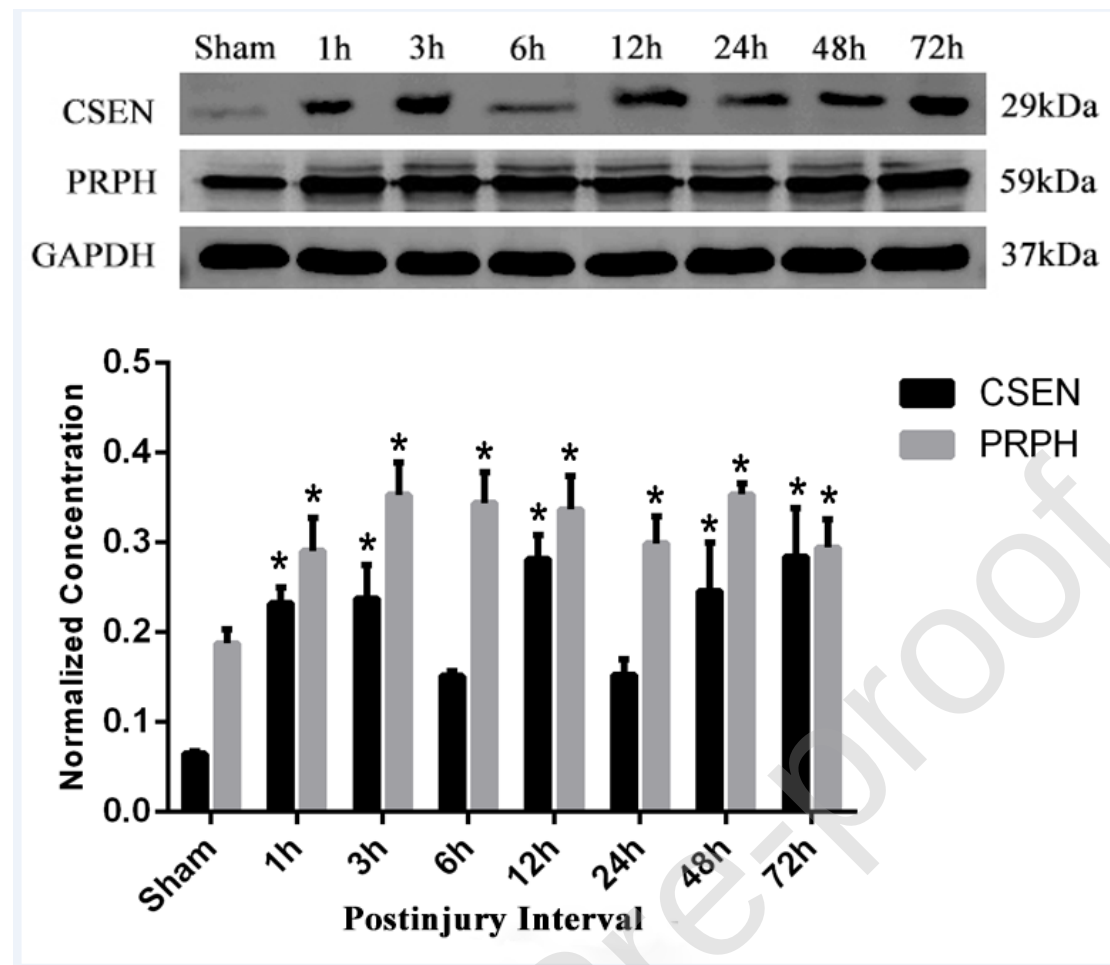


Fig. 8 Western blotting analysis in the corpus callosum. Peripherin (PRPH) and calsenilin (CSEN) were examined in triplicate and normalized to GAPDH levels (loading control) for quantitative analysis (*: there were significant differences between the injury groups and the sham group, $p < 0.05$).

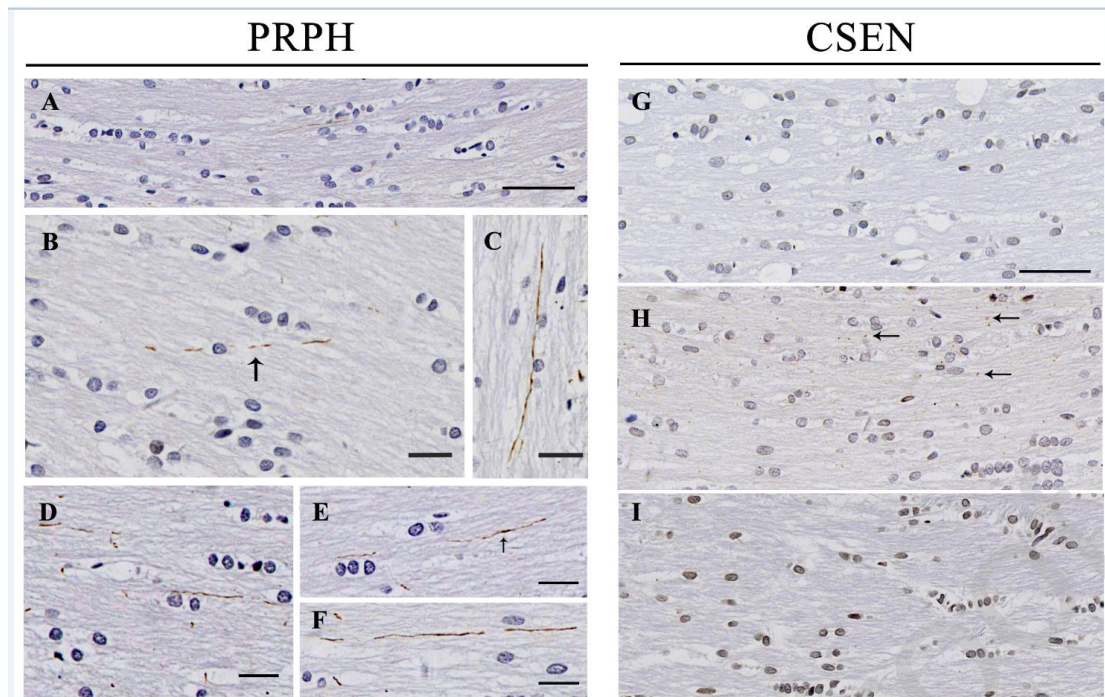


Fig. 9 A representative example of IHC staining showing PRPH in the corpus callosum (A-F). A. A single axon reactive for PRPH of a sham rat with normal morphology (bar = 50 μ m); B. PRPH positive profiles in the corpus callosum 6 h postinjury. Note that a single axon reactive for PRPH (black arrow) revealed a patchy immunoreactivity likely indicative of axonal disconnection (bar = 20 μ m); C. A single PRPH positive axon at 6 h postinjury with some mild swellings and beaded morphology (bar = 20 μ m); D. Multiple PRPH positive profiles at 48 h postinjury displaying a variety of injured morphologies including undulating, beading and mild swelling (bar = 20 μ m); E. A single axon reactive for PRPH (black arrow) 48 h postinjury with undulating and mild swelling morphologies (bar = 20 μ m); F. An axon with incoherent immunoreactivity for PRPH 72 h postinjury showing swelling and undulating morphology indicative of axonal disconnection and degeneration (bar = 20 μ m).

A representative example of IHC staining showing PRPH in the corpus callosum (G-I). G. Sham rats displayed a low CSEN immunoreactivity in neuroglial cells; H. 1h postinjury, CSEN immunoreactivity was more intense and punctate CSEN-reactive distributions were observed in some axons (black arrow). I. CSEN positive neuroglial cells were more prominent at 72h postinjury (All scale bar = 50 μ m).

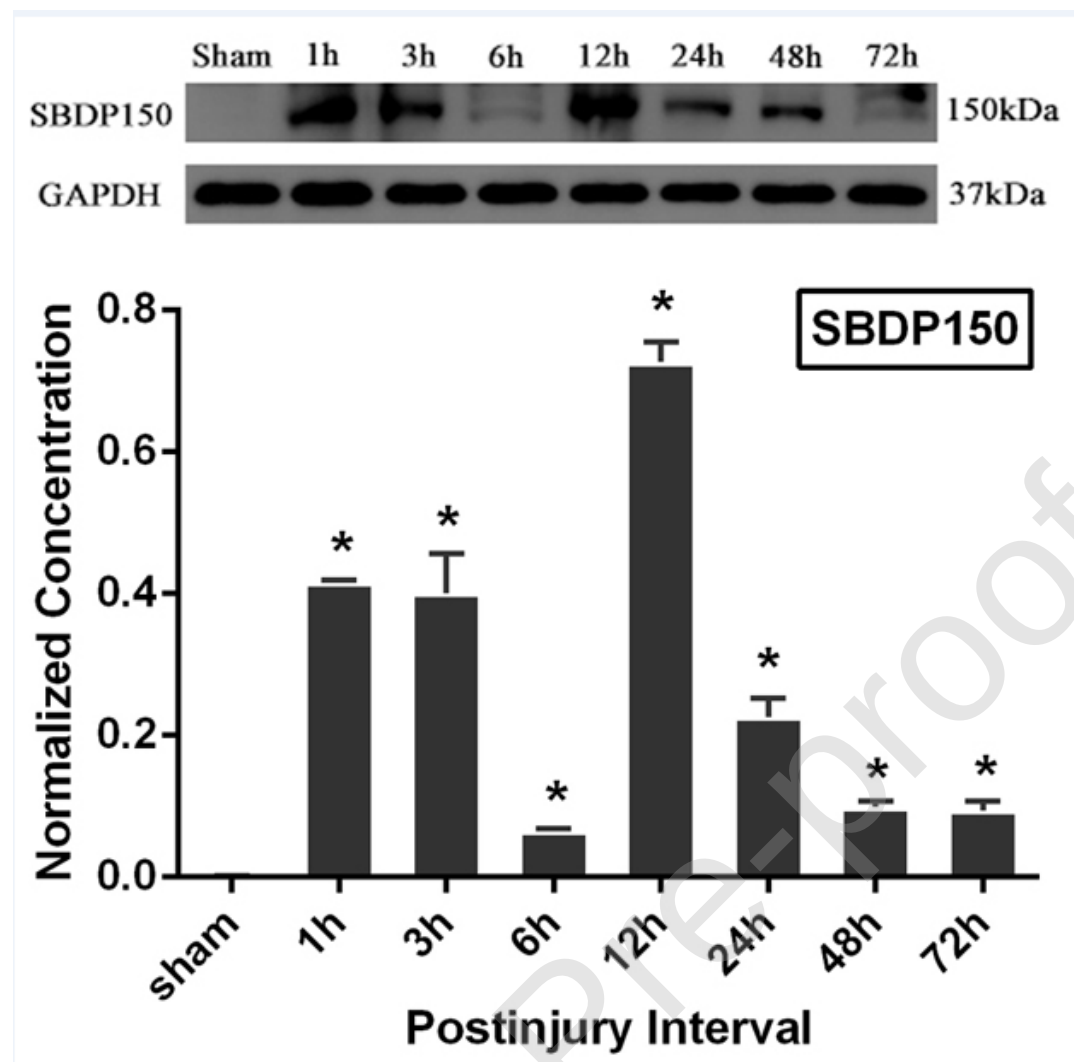


Fig. 10 Western blotting analysis in corpus callosum. SBDP150 was examined in triplicate and normalized to GAPDH levels (loading control) for quantitative analysis (*: there were significant differences between the injury group and the sham group, $p < 0.05$).

Table 1: The number of rats used for different experiments

Application	Number of rats (total/ each group)	
	DAI groups	Sham group
iTRAQ	70/ 10	10/ 10
Histopathological examination	35/ 5	5/ 5
Western blotting	21/ 3	3/ 3

Table 2: Common differential proteins and their fold changes at all seven time points post-DAI

Accession ^a	Protein name	Gene symbol	DAI/sham (fold change)						
			1 h-post injury	3 h-post injury	6 h-post injury	12 h-post injury	24 h-post injury	48 h-post injury	72 h-post injury
D3ZUL3	Collagen type VI alpha 1 chain	Col6a1	0.325	0.461	0.151	0.261	0.497	0.380	0.470
Q4KM74	Vesicle-trafficking protein SEC22b	Sec22b	0.286	0.261	0.370	0.310	0.242	0.470	0.139
P04177	Tyrosine 3-monooxygenase	Th	0.363	0.356	0.283	0.179	0.437	0.449	0.207
F1LNM6	Sodium- and chloride-dependent glycine transporter 2	Slc6a5	0.316	0.483	0.466	0.402	0.384	0.421	0.36
P10687	1-phosphatidylinositol 4,5-bisphosphate phosphodiesterase beta-1	Plcb1	2.249	3.105	3.311	3.802	3.532	3.373	3.192
D4AEI5	Myelin expression factor 2	Myef2	3.162	2.051	2.858	3.020	2.051	2.655	2.831
M0RCA6	Advillin	Avil	5.495	5.445	6.081	5.649	3.342	5.546	6.252
P09215	Protein kinase C	Prkcd	4.571	2.355	4.055	5.105	4.831	3.908	4.055
D4A4P8	Protein tweety homolog	Ttyh2	5.649	5.012	4.529	3.436	4.018	5.105	3.499

D4A6U 1	A disintegrin and metallopeptidase domain 11 (Predicted), isoform CRA_a	Adam 11	6.36 8	3.342	4.52 9	3.467	2.228	4.406	5.346
F1M324	Phosphoinositide phospholipase C	Plcl2	6.98 2	3.221	4.05 5	3.802	4.018	2.858	3.631
D4AE58	KN motif and ankyrin repeat domains 1	Kank1	5.49 5	17.378	4.48 7	6.368	6.918	17.86 5	4.656
P21807	Peripherin	Prph	8.71 0	9.036	9.37 6	8.166	3.945	10.37 5	7.586
Q9JM47	Calsenilin	Kcnip 3	10.5 68	4.055	9.20 4	7.447	9.376	10.28 0	3.767

^a Database accession numbers; ^b UniProt Database cannot find this protein, so this accession number is found in RGD (Rat Genome Database).

Table 3: List of differential proteins with different biological processes and their fold changes in DAI groups.

Accession ^a	Protein name	Gene symbol	DAI/sham (fold change)						
			1 h-post injury	3 h-post injury	6 h-post injury	12 h-post injury	24 h-post injury	48 h-post injury	72 h-post injury
Calcium ion regulation									
P10687	1-phosphatidylinositol 4,5-bisphosphate phosphodiesterase beta-1	Plcb1	2.249	3.105	3.311	3.802	3.532	3.373	3.192
B0LPN4	Ryanodine receptor 2	Ryr2	3.133	2.831	3.251	2.965	3.767	3.105	/
P09215	Protein kinase C delta type	Prkcd	4.571	2.355	4.055	5.105	4.831	3.908	4.055
Q5EB C0	Inter alpha-trypsin inhibitor, heavy chain 4	Itih4	2.831	3.631	2.466	/	/	/	2.27
P48199	C-reactive protein	Crp	2.443	4.487	2.858	/	2.679	2.63	2.965
D3ZU64	VPS37B, ESCRT-I subunit	Vps37b	2.128	2.754	2.884	2.355	0.429	2.312	/
Q02563	Synaptic vesicle glycoprotein 2A	Sv2a	/	/	2.443	3.048	2.355	/	/
Q561K2	NEL-like 2 (Chicken)	Nell2	3.945	/	/	/	/	/	/
Q6IN33	Calciopressin-1	Rcan1	3.837	/	4.207	3.837	/	3.281	4.487
Q9JM47	Calsenilin	Kcni p3	10.568	4.055	9.204	7.447	9.376	10.28	3.767
F1MAD3	Polycystin 1, transient receptor potential channel-interacting	Pkd1	2.249	/	3.342	/	/	2.679	/
P16086	Spectrin alpha chain, non-erythrocytic 1	Sptan1	0.457	/	/	/	/	/	/

Q9Z1 P2	Alpha-actinin-1	Actn 1	0.45 3	/	/	/	/	/	/
P4803 7	Annexin A6	Anxa 6	0.40 2	0.21 7	/	/	0.42 5	0.49 7	/
P0DP 29	Calmodulin-1	Calm 1	0.38 7	/	/	/	/	2.14 8	/
P1150 5	Plasma membrane calcium- transporting ATPase 1	Atp2 b1	0.42 5	/	/	/	/	/	/
Q635 37	Synapsin-2	Syn2	0.39 1	/	/	/	/	/	/
Q630 81	Protein disulfide- isomerase A6	Pdia 6	0.46 1	/	/	/	/	/	/
P4772 8	Calretinin	Calb 2	0.32 8	/	0.44 5	/	0.40 9	/	0.34
P1841 8	Calreticulin	Calr	0.29 4	/	0.48 3	/	0.47	/	/
P6276 2	Visinin-like protein 1	Vsnl 1	0.46 1	/	/	/	/	/	/
Q641 19	Myosin light polypeptide 6	Myl6	0.45 7	/	/	/	/	/	/
P6310 0	Calcineurin subunit B type 1	Ppp3 r1	0.26 3	/	/	/	/	/	/
F1M AB9	Tumor protein D52	Tpd5 2	0.36 6	/	/	/	0.49 7	/	/
P0262 5	Parvalbumin alpha	Pval b	0.37 7	/	/	0.20 7	0.22 5	/	0.13 8
P6302 9	Translationally- controlled tumor protein	Tpt1	0.13 6	/	0.41 7	/	0.42 1	/	/
Q641 Z8	Peflin	Pef1	0.40 2	/	/	0.28 8	0.10 6	/	0.15
P3285 1	Syntaxin-1A	Stx1 a	/	2.33 3	/	3.13 3	/	2.80 5	/
Q5U2 X6	Coiled-coil domain- containing protein 47	Ccdc 47	/	2.16 8	2.24 9	/	/	2.39 9	/
D4A BT0 (P592 15)	Guanine nucleotide- binding protein	Gnao 1	/	/	/	2.42 1	/	/	/

	G(o) subunit alpha								
D3ZF54	Anoctamin	Ano10	/	0.24	/	/	/	/	/
Q99JE6	1-phosphatidylinositol 4,5-bisphosphate phosphodiesterase beta-3	Plcb3	/	0.377	/	/	/	/	/
D3ZUB0	Reticulocalbin 1	Rcn1	/	0.268	/	/	/	/	/
Q1HAQ0	Lysophosphatidylcholine acyltransferase 1	Lpcat1	/	0.402	10.965	19.953	/	14.06	7.516
F1LR98	Cadherin 10	Cdh10	/	/	2.208	0.398	/	/	/
P54287	Voltage-dependent L-type calcium channel subunit beta-3	Cacnb3	/	/	2.377	/	/	/	/
D3ZPY1 (A9LNM8)	cGMP-dependent protein kinase	Prkg1	/	/	2.051	/	/	/	2.188
G3V9M6	Fibrillin 1	Fbn1	/	/	0.31	/	/	0.453	/
M0R979	Thrombospondin 1	Thbs1	/	/	0.409	/	0.453	0.203	/
D3ZYI0	Grancalcin	Gca	/	/	0.461	/	/	/	/
P05964	Protein S100-A6	S100a6	/	/	0.377	2.858	/	/	/
B0BMS8	Myl9 protein	Myl9	/	/	/	0.492	/	/	/
P29994	Inositol 1,4,5-trisphosphate receptor type 1	Itpr1	/	/	/	/	2.228	/	/
D3ZG78	Zinc finger ZZ-type and EF-hand domain-containing 1	Zzef1	/	/	/	/	2.168	/	/

Q8K3 P6	Calcium-binding mitochondrial carrier protein SCaMC-2	Slc2 5a25	/	0.2	/	/	0.27 8	0.47	/
P1127 5	Calcium/calmodu lin-dependent protein kinase type II subunit alpha	Cam k2a	/	/	/	/	/	2.46 6	/
G3V6 M3	Synaptotagmin II	Syt2	/	0.33 4	/	/	/	0.41 3	/
P0260 0	Myosin light chain 1/3, skeletal muscle isoform	Myl1	/	/	/	/	/	/	0.44 1
P5431 1	Guanine nucleotide- binding protein G(I)/G(S)/G(T) subunit beta-1	Gnb1	/	0.22 9	/	/	/	/	/
Q9Z2 L0	Voltage- dependent anion- selective channel protein 1	Vdac 1	/	0.12 6	/	/	/	/	/
P2071 7	Protein-arginine deiminase type-2	Padi 2	/	0.41 7	/	/	/	/	/
P5431 3	Guanine nucleotide- binding protein G(I)/G(S)/G(T) subunit beta-2	Gnb2	/	0.09 9	/	/	/	/	/
Q079 36	Annexin A2	Anxa 2	/	0.18 5	/	/	/	/	/
Q5U3 62	Annexin	Anxa 4	/	0.10 8	/	/	/	/	/
P0715 0	Annexin A1	Anxa 1	/	0.45 7	/	/	/	/	/
Q626 36	Ras-related protein Rap-1b	Rap1 b	0.18 9	/	/	/	/	/	/
Cytoskeleton organization									
P2180 7	Peripherin	Prph	8.71 0	9.03 6	9.37 6	8.16 6	3.94 5	10.3 75	7.58 6
M0R CA6	Advillin	Avil	5.49 5	5.44 5	6.08 1	5.64 9	3.34 2	5.54 6	6.25 2

Q6IF U9	Keratin 16	Krt1 6	3.66 4	/	7.94 3	/	/	/	/
Q9JI9 2	Syntenin-1	Sdcb p	/	2.27 0	2.14 8	2.48 9	2.72 9	2.48 9	/
P3197 7	Ezrin	Ezr	/	/	2.10 9	/	/	/	/
P9784 6	Contactin-associated protein 1	Cntn ap1	/	/	/	/	/	/	2.14 8
P1608 6	Spectrin alpha chain, non-erythrocytic 1	Spta n1	0.45 7	/	/	/	/	/	/
Q6P9 V9	Tubulin alpha-1B chain	Tuba 1b	0.16 1	/	/	/	0.09 5	/	0.47
P1952 7	Neurofilament light polypeptide	Nefl	0.31 9	/	0.45 3	0.27 5	0.29 6	0.35	0.27 5
P1283 9	Neurofilament medium polypeptide	Nef m	0.27 8	/	/	0.34	0.29 6	/	0.36
F1LR Z7	Neurofilament heavy polypeptide	Nefh	0.43 3	/	/	0.43 3	0.40 2	/	0.46 6
P0479 7	Glyceraldehyde-3-phosphate dehydrogenase	Gapd h	0.35 3	/	/	/	/	/	/
Q9Z0 W5	Protein kinase C and casein kinase substrate in neurons protein 1	Pacsi n1	0.42 1	/	/	/	/	/	/
A0A0 G2K1 32	Formin-like 2	Fmnl 2	0.46 6	/	/	/	/	/	/
D4A4 L4	MAP7 domain-containing 2	Map 7d2	0.40 6	/	/	/	/	/	0.49 2
P3492 6	Microtubule-associated protein 1A	Map 1a	0.45 7	/	/	/	/	0.47	/
D3ZZ P8	Cullin 9	Cul9	0.46 1	/	/	/	/	/	/
P0183 0	Thy-1 membrane glycoprotein	Thy1	/	0.32 5	/	/	/	0.36 3	/
D3ZH G3	Protein tyrosine kinase 7	Ptk7	/	0.46 6	/	0.36 3	/	/	/

C0JP T7	Filamin A	Flna	/	/	0.13 4	0.30 2	/	/	/
Q5PQ P4	Cdc42 effector protein 2	Cdc4 2ep2	/	/	/	/	0.26 8	/	/
F1M7 75	Diaphanous- related formin 1	Diap h1	/	/	/	/	/	/	0.41 3
F1LM V9	Coronin	Coro 2b	/	0.45 3	/	/	/	/	/
D3ZD U5	Profilin	Pfn2	/	0.44 1	/	/	/	/	/
Apoptotic process									
P0277 0	Serum albumin	Alb	2.85 8	4.28 5	/	/	/	0.36	/
P0921 5	Protein kinase C delta type	Prkc d	4.57 1	2.35 5	4.05 5	5.10 5	4.83 1	3.90 8	4.05 5
Q9JM 47	Calsenilin	Kcni p3	10.5 68	4.05 5	9.20 4	7.44 7	9.37 6	10.2 8	3.76 7
P4646 2	Transitional endoplasmic reticulum ATPase	Vcp	0.47 4	/	/	/	/	/	/
P0479 7	Glyceraldehyde- 3-phosphate dehydrogenase	Gapd h	0.35 3	/	/	/	/	/	/
P4803 7	Annexin A6	Anxa 6	0.40 2	0.21 7	/	/	0.42 5	0.49 7	/
P0911 7	Fructose- bisphosphate aldolase C	Aldo c	0.47 9	/	/	/	0.44 5	/	/
Q66H F1	NADH- ubiquinone oxidoreductase 75 kDa subunit, mitochondrial	Nduf s1	0.36	/	/	/	/	/	0.47
P6303 9	60 kDa heat shock protein, mitochondrial	Hspd 1	0.18 5	/	/	/	0.42 9	/	/
P0676 1	78 kDa glucose- regulated protein	Hspa 5	0.31	/	/	/	0.47 4	/	/
A0A0 96MJ N4	Protein Sept4	4- Sep	0.47 4	/	/	/	/	/	/
Q6P1 36	Hyou1 protein	Hyou 1	0.45 7	/	/	/	/	/	/

P04785	Protein disulfide-isomerase	P4hb	0.363	/	/	/	/	/	/
Q63081	Protein disulfide-isomerase A6	Pdia6	0.461	/	/	/	/	/	/
F1LVR4	Calcium/calmodulin-dependent protein kinase ID	Camk1d	0.492	/	/	/	/	/	/
Q9WVR7	Protein phosphatase 1F	Ppm1f	0.308	/	/	/	0.247	/	/
P51647	Retinal dehydrogenase 1	Aldh1a1	0.387	0.483	/	0.441	/	/	0.479
D4A5L9	Similar to Cytochrome c, somatic	LOC679794	0.353	/	/	/	0.445	/	/
P63086	Mitogen-activated protein kinase 1	Mapk1	0.466	/	/	/	/	/	/
P06907	Myelin protein P0	Mpz	0.137	0.221	0.217	/	0.377	0.325	0.217
P23928	Alpha-crystallin B chain	Cryab	0.483	/	/	/	/	/	/
P60905	DnaJ homolog subfamily C member 5	Dnajc5	0.479	/	/	/	/	/	/
Q04631	Protein farnesyltransferase/geranylgeranyltransferase type-1 subunit alpha	Fnta	0.305	/	/	/	/	/	/
P63029	Translationally-controlled tumor protein	Tpt1	0.136	/	0.417	/	0.421	/	/
Q9WVJ6	Tissue-type transglutaminase	Tgm2	0.488	/	/	0.474	/	/	/
F1LM55	Cell cycle and apoptosis regulator 2	Ccar2	/	2.07	/	/	/	/	/
P49911	Acidic leucine-rich nuclear phosphoprotein 32 family member A	Anp32a	/	2.559	2.07	2.78	/	/	2.333
P14480	Fibrinogen beta chain	Fgb	/	2.051	/	/	/	/	3.133

P01830	Thy-1 membrane glycoprotein	Thy1	/	0.325	/	/	/	0.363	/
D3Z9G0	Tnf receptor-associated factor 3	Traf3	/	0.316	/	/	/	/	/
Q10758	Keratin, type II cytoskeletal 8	Krt8	/	/	2.858	/	/	/	0.421
Q569C9	Golgi phosphoprotein 3	Golp h3	/	/	2.188	/	/	/	/
C0JPT7	Filamin A	Flna	/	/	0.134	0.302	/	/	/
P04937	Fibronectin	Fn1	/	/	0.209	/	/	/	/
M0R979	Thrombospondin 1	Thbs1	/	/	0.409	/	0.453	0.203	/
Q8CHN6	Sphingosine-1-phosphate lyase 1	Sgpl1	/	/	0.394	0.081	0.174	/	/
P07722	Myelin-associated glycoprotein	Mag	/	/	/	0.466	/	/	/
P21708	Mitogen-activated protein kinase 3	Map k3	/	/	/	0.429	/	/	0.466
O35331	Pyridoxal kinase	Pdxk	/	/	/	0.461	/	/	/
P01355	Cholecystokinin	Cck	/	/	/	/	2.704	/	/
Q5XIF5	Sphingolipid delta(4)-desaturase DES1	Degs1	/	/	/	/	3.02	/	/
F1M9D6	Signal transducer and activator of transcription	Stat1	/	/	/	/	0.492	/	/
P11762	Galectin-1	Lgals1	/	/	/	/	0.466	/	/
H9KVF3	Serine/threonine-protein kinase 24	Stk24	/	/	/	/	0.492	/	/
Q5BJY9	Keratin, type I cytoskeletal 18	Krt18	/	/	/	/	0.483	/	/
Q498U6	DNAation factor, alpha subunit	Dffa	/	/	/	/	0.133	/	0.215

P0464 2	L-lactate dehydrogenase A chain	Ldha	/	/	/	/	/	2.33 3	2.27
G3V9 13	Heat shock 27kDa protein 1	Hspb1	/	/	/	/	/	2.46 6	3.22 1
Q6RJ R6	Reticulon-3	Rtn3	/	/	2.10 9	/	/	/	/
P0570 8	Hexokinase-1	Hk1	/	0.46 1	/	/	/	/	/
Q9Z2 L0	Voltage-dependent anion-selective channel protein 1	Vdac1	/	0.12 6	/	/	/	/	/
P8115 5	Voltage-dependent anion-selective channel protein 2	Vdac2	/	0.24 2	/	/	/	/	/
P6324 5	Receptor of activated protein C kinase 1	Rack1	/	0.46 6	/	/	/	/	/
B0BN 52	Mitochondrial carrier 2	Mtch2	/	0.48 8	/	/	/	/	/
Q9W UD9	Proto-oncogene tyrosine-protein kinase Src	Src	/	0.24 7	/	/	/	/	/
Q642 44	ADP-ribosyl cyclase/cyclic ADP-ribose hydrolase 1	Cd38	/	0.42 1	/	/	/	/	/
P0715 0	Annexin A1	Anxa1	/	0.45 7	/	/	/	/	/
Inflammatory response									
M0R BJ7	Complement C3	C3	2.16 8	3.07 6	/	/	/	/	/
P1747 5	Alpha-1-antiproteinase	Serpin1	2.24 9	5.2	2.78	/	/	/	/
P2409 0	Alpha-2-HS-glycoprotein	Ahsg	2.24 9	4.20 7	2.27	/	/	/	/
Q6IF U9	Keratin 16	Krt16	3.66 4	/	7.94 3	/	/	/	/
P6303 9	60 kDa heat shock protein, mitochondrial	Hspd1	0.18 5	/	/	/	0.42 9	/	/

P60203	Myelin proteolipid protein	Plp1	0.47	0.21	0.291	/	/	/	/
Q6IMF3	Keratin, type II cytoskeletal 1	Krt1	0.203	0.394		0.466		0.356	0.185
Q9WVJ6	Tissue-type transglutaminase	Tgm2	0.488	/	/	0.474	/	/	/
P04638	Apolipoprotein A-II	Apoa2	/	2.489	/	/	/	/	/
P06866	Haptoglobin	Hp	/	0.429	0.497	/	/	/	0.492
M0R979	Thrombospondin 1	Thbs1	/	/	0.409	/	0.453	0.203	/
Q5XIM7	Lysine--tRNA ligase	Kars	/	/	/	0.492	/	/	/
P20291	Arachidonate 5-lipoxygenase-activating protein	Alox5ap	/	/	/	0.194	/	/	0.092
P80254	D-dopachrome decarboxylase	Ddt	/	/	/	/	0.483	/	0.347
P09006	Serine protease inhibitor A3N	Serpi na3n	/	/	/	/	/	/	2.07
P17256	Mevalonate kinase	Mvk	/	/	/	/	/	/	0.449
F1LM10	Protein kinase C theta type	Prkc q	/	/	/	/	/	/	0.328
P10760	Adenosylhomocysteinase	Ahcy	/	0.413	/	/	/	/	/
P07150	Annexin A1	Anxa1	/	0.457	/	/	/	/	/
D3ZQP6	Sema domain, immunoglobulin domain (Ig), and GPI membrane anchor, (Semaphorin) 7A (Predicted)	Sema7a	/	0.38	/	/	/	/	/
P06399	Fibrinogen alpha chain	Fga	7.87	5.649	4.093	/	6.026	3.499	10.471
Q5EBC0	Inter alpha-trypsin inhibitor, heavy chain 4	Itih4	2.831	3.631	2.466	/	/	/	2.27
P48199	C-reactive protein	Crp	2.443	4.487	2.858	/	2.679	2.63	2.965

Q6A YT1	Similar to 9230105E10Rik protein	Trim 5	0.48 3	0.41 3	/	/	/	/	/
P1448 0	Fibrinogen beta chain	Fgb	/	2.05 1	/	/	/	/	3.13 3
D3Z9 G0	Tnf receptor- associated factor 3	Traf3	/	0.31 6	/	/	/	/	/
Q4KL Z6	Bifunctional ATP-dependent dihydroxyaceton e kinase/FAD- AMP lyase (cyclizing)	Tkfc	/	/	0.44 5	/	/	/	/
D3ZE V8	Sushi domain containing 2 (Predicted), isoform CRA_a	Susd 2	/	/	/	/	/	/	2.08 9
B5DF 13	RCG63400	Trim 59	/	/	/	2.53 5	/	/	0.29 9
P8597 3	Purine nucleoside phosphorylase	Pnp	/	0.34 7	/	/	/	/	/
P0268 0	Fibrinogen gamma chain	Fgg	2.27 0	2.24 9	/	/	2.10 9	2.10 9	3.90 8
Q6P9 V9	Tubulin alpha-1B chain	Tuba 1b	0.16 1	/	/	/	0.09 5	/	0.47
P1278 5	Fatty acid synthase	Fasn	0.48 8	0.40 9	/	/	/	/	/
P0268 8	Myelin basic protein	Mbp	0.24 9	/	0.21 1	0.40 9	0.16 3	0.47 4	0.35
Q5U3 62	Annexin	Anxa 4	/	0.10 8	/	/	/	/	/